An Explainable Hybrid Deep Learning Approach for Multi-Class Brain Disease Detection Using MRI Images

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CERTIFICATE

This is to certify that this project report titled **"An Explainable Hybrid Deep Learning Approach for Multi-Class Brain Disease Detection Using MRI Images"** is a bonafide record of work done by **RAHUL PUDI (228W1A1251), PAVAN KUMAR KOLLIPATI (228W1A1228)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2025-26.

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PROJECT SUMMARY

S.No	Item	Description
1	**Project Title**	**An Explainable Hybrid Deep Learning Approach for Multi-Class Brain Disease Detection Using MRI Images**
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4	**Research Group**	**Deep Learning, Medical Image Analysis, Neuroimaging**
5	**Application Area**	**Healthcare**
6	**Aim of the Project**	**To develop a hybrid deep learning-based multi-class brain disease detection system using MRI images by integrating segmentation, feature extraction, and machine learning techniques, along with Grad-CAM for visual explainability.**
7	**Project Outcomes**	• **Accurate classification of brain tumor, Alzheimer's disease, and multiple sclerosis** • **Improved diagnostic support using hybrid models with explainable results**

Student Signatures

1.
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Signature of the Guide

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LIST OF EQUATIONS

Equation	Brief Description
(1)	Normalisation
(2)	Feature Extraction
(3)	Flatten
(4)	Linear Transformation
(5)	ReLu Activation Function
(6)	DropOut
(7)	Softmax Function
(8)	Categorical Cross Entropy Loss
(9)	Admax Optimizer
(10)	Learning rate
(11)	Normalisation
(12)	Compound Scaling
(13)	Flatten
(14)	Global pooling
(15)	Batch Normalisation
(16)	Fully Connected layers
(17)	Relu Activation Function
(18)	DropOut
(19)	Softmax Function
(20)	Categorical Cross Entropy Loss
(21)	Adaptive Optimizer

(22)	Convolution Operator
(23)	Relu Activation Function
(24)	Pooling
(25)	Flatten
(26)	Fully Connected Layers
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ABSTRACT

With many individuals being diagnosed with neurological conditions like brain tumors, Alzheimer's disease and multiple sclerosis, a reliable and expedient means of diagnosing these conditions is essential. Traditional diagnostic methods for brain imaging are usually constrained to the manual interpretation of MRI scans which can be time intensive as well as subjective. To aid clinicians in this process, a hybrid deep learning model has been developed to classify brain MRI images for the three conditions combined using segmentation, feature extraction, and a machine learning classification algorithm. The diagnostic model employs three novel methods to evaluate images – U-Net++, EfficientNetV2-S and XGBoost, which will allow segmentation and context to the regions of interest (ROI) in order to improve the classification performance of each image through improved feature extraction. Along with this model, a second comparative diagnostic model was developed to compare the diagnostics of the same MRI images using three different methods, U-Net, DenseNet201, and LightGBM. The brains images from the brain tumors, Alzheimer's disease, and multiple sclerosis categories were combined to create one training set with an equal number of representative images for each condition to ensure uniform learning. Using the respective MRI images, the proposed hybrid model was shown to provide superior classification of the brain images with an accuracy of 91%. In addition, Grad-CAM was used on the hybrid model to help provide transparency regarding the classification of the respective images by providing an ability to identify those regions that contributed most to making a classification decision. Overall, these results demonstrate that the proposed hybrid model provides an increase in both classification accuracy and reliability and has the potential to be implemented in clinical decision support systems.

Keywords: Brain MRI Classification, Brain Tumor Detection, Alzheimer's Disease Detection, Multiple Sclerosis Detection, Hybrid Deep Learning, U-Net++, EfficientNetV2-S, XGBoost, Medical Image Segmentation, Feature Extraction, Multi-Class Classification, Grad-CAM, Computer Vision, Deep Learning, Clinical Decision Support.

Chapter-1

Introduction

1.1 Origin of the Problem

Brain tumors, Alzheimer's disease and multiple sclerosis are neurological disorders that have become a significant health issue globally owing to their rising occurrence and devastating effects on the lives of mankind. Brain tumors are also one of the life threatening conditions that need prompt and precise diagnosis to enable proper planning of treatment [1]. Alzheimer disease is a progressive neurodegenerative condition that causes the loss of memory and cognitive impairment that presents a huge burden to both patients and the health care system [2]. Multiple sclerosis is a long-term neurological disease that can impact the central nervous system and damage nerve fibers, which results in physical and mental disabilities [3].

The diagnoses of these neurological disorders are often done using Magnetic Resonance Imaging (MRI) because this test is high resolution and non-invasive. Nevertheless, the process of manual examination of the MRI scans by the radiologists is time consuming, labor intensive and subject to human error and inter-observer variability [2]. In multiple sclerosis, the diagnosis is especially complicated, as there is no specific diagnostic test, and it depends on clinical experience in conjunction with the use of the imaging methods [4]. On the same note, the initial stages of Alzheimer disease are hard to diagnose since the alterations in the brain structure are very subtle and can hardly be distinguished by the conventional method [5].

Despite the encouraging outcomes of deep learning methods, notably Convolutional Neural Networks (CNNs), the models used to date have certain drawbacks. Most methods are specific to detecting a single disease and cannot be generalized between two or more neurological diseases [6]. Furthermore, tumor heterogeneity, differences in lesion size, shape, and intensity, and intricate brain structures render it hard to classify them using conventional models [7]. The other issue is that there is a scarcity of large annotated medical datasets, which impacts the performance and generalization potential of deep learning models [8].

Moreover, conventional machine learning algorithms are based on handcrafted features, which cannot be relied on to represent complicated patterns in MRI images. Even state-of-the-art deep learning models tend to be not very robust and interpretable, which restricts their utility in clinical practice. The growing variety in the nature of diseases also makes classification more complex, with models failing to differentiate between the most related neurological disorders [9].

Thus, the roots of this issue are in the inefficiency of the manual diagnosis method, the ineffectiveness of the single-disease identification systems, and the inability of the current solutions to cope with the multi-disease classification. Such difficulties indicate the necessity to use a sophisticated hybrid model to combine segmentation, feature extractions, and machine learning to enhance diagnostic accuracy, robustness, and clinical reliability.

1.2 Basic definitions and Background

Brain Tumor: A brain tumor is an overgrowth of cells in the brain, which may be harmless or cancerous. The tumors are of different sizes, shapes and intensities, and their detection and categorization is a difficult task. Proper identification is critical in planning treatment and enhancing patient survival rates [2]. Nevertheless, the complexity of structures and heterogeneity of tumors tends to decrease the efficiency of conventional diagnostic techniques [3].

Alzheimer's Disease: Alzheimer is a progressive neurodegenerative disorder that majorly involves memory, ability to think and behavior. It is the most prevalent dementia cause in the world and is a slow process. Early diagnosis is essential, and intervention can delay the disease process and enhance patient outcomes [7]. MRI is used in the structural changes like brain atrophy and degeneration of tissues related to the Alzheimer disease [8].

Multiple Sclerosis (MS): Multiple Sclerosis (MS) is a disease that will effect the brain and spinal cord of our body. MRI has become widespread in detection of lesions and monitoring of the disease progression but it is not easily diagnosticated because there is no clear-cut test .

Deep Learning: Deep learning is a branch of artificial intelligence that applies multi-layered neural networks to automatically derive patterns on big data sets. It has greatly enhanced medical image analysis performance through automated extractions and classification of features [9].

Convolutional Neural Networks (CNNs): Convolutional Neural Networks are the models used for classification of the images. They extract hierarchical features like edges, textures and shapes automatically in MRI images, and thus they are very useful in detecting and classifying diseases [10]. The CNN-based models have proven to be superior to the traditional machine learning methods.

Image Segmentation: Image segmentation involves breaking down an image into significant portions with a view of isolating significant structures. Segmentation is applied in medical imaging to identify the region of interest (ROI) in MRI images e.g. tumors or lesions. Accurate segmentation of medical images is a common technique, such as U-Net and its variations [11].

Feature Extraction: The process of extracting features is to convert raw image data into representations that can be used to classify the image. EfficientNet and DenseNet are deep learning models, which are most likely to extract high-level features of MRI images, leading to the achievement of better classification performance [12].

Machine Learning Classifiers: Once the features are extracted, the machine learning algorithms are applied in classification. Gradient boosting and ensemble learning are techniques that enhance the accuracy of predictions by using a combination of weak learners to become a strong model. These methods have been found to be particularly useful in processing complicated medical data [13].

Explainable AI (XAI): Explainable Artificial Intelligence (XAI) is the methods used to explain deep learning models. Techniques like Grad-CAM can be used to visualize what parts of an image determine its predictions to the model and can make it more trustworthy and transparent in clinical practice [14].

Hybrid Deep Learning Models: Hybrid models are models that fuse several methods like segmentation, deep learning and machine learning to improve performance. These

types of models exploit the advantages of every element, including ROI extraction, feature learning, and classification, to enhance the accuracy of the diagnostic in medical imaging tasks [20].

Development of medical imaging machines, especially MRI, has made a great contribution in the diagnosis of neurological conditions. Nonetheless, the interpretation of MRI scans manually is not an easy task because of the intricacy of the brain structures and lack of uniformity in disease patterns. It is laborious and subject to human fallacy, particularly where there are minor abnormalities or when there is a disease at its early stages [2]. Other neurological disorders like brain tumors, Alzheimer disease and multiple sclerosis are challenging to diagnose because of the difference in size, shape and intensity of the lesion. Also, the detection at an early stage is challenging owing to the fact that the alterations in the brain structure are usually slight and cannot be easily recognized by the conventional means [4]. The most recent advances in artificial intelligence, especially in deep learning have transformed the field of medical image analysis. Methods like CNNs have shown high-levels of ability to automatically extract features and enhance the classification accuracy [6]. Nevertheless, current models tend to concentrate on detecting a single disease and cannot be generalized among various neurological disorders [7]. The other shortcoming of traditional methods is the inefficiency in isolating the area of interest (ROI), which results in loss of accuracy with background information that is not relevant. Moreover, deep learning models need big annotated datasets, which can be scarce in the medical sector, impacting model performance and generalization [10]. Hybrid methods involving segmentation, deep learning, and machine learning methods have been suggested to overcome these difficulties. These techniques are used to enhance accuracy by paying attention to the important areas in the image, generating meaningful features and increasing the performance of the classifier.

1.3 Problem Statement with Objectives and Outcomes

1.3.1 Problem Statement

Medical diagnosis of neurological disorders like brain tumors, Alzheimer and multiple sclerosis poses a major challenge to medical diagnosis because they are complex,

variable and progressive. It is necessary to detect these diseases earlier and accurately in order to plan the treatment and enhance the survival rates of patients. Nonetheless, traditional diagnostic procedures are based on hand interpretation of MRI scans, which is time-consuming, subjective, and requires the skills of medical personnel.

The majority of current systems are developed to classify a single disease and do not incorporate machine learning and deep learning methods to address numerous neurological disorders in a single framework, even though the introduction of machine learning and deep learning has allowed automating the diagnostic process. Moreover, most of the models do not emphasize on the region of interest (ROI), and therefore, unnecessary background information is included, which adversely impacts classification results.

Moreover, due to variations in the disease characteristics like heterogeneity of tumors, the pattern of lesions and microscopic alterations in the initial stages of the diseases, classification of diseases has proven to be challenging. Overfitting, inability to generalize, and poor interpretability are other issues that deep learning models have, and it complicates their application to clinical practice.

Thus, there is a necessity to create a powerful, precise, and interpretable hybrid a combination of segmentation, deep learning, and machine learning to classify multi-disease effectively based on brain MRI images.

1.3.2 Objectives

The primary aim of the project is to create a smart medical diagnosis support system, which will be able to automatically identify and categorize various brain diseases using MRI images and give results that can be interpreted. The project will focus on integrating medical image segmentation, use of deep learning to produce feature and machine learning to classify them such that the disease prediction and visual explanation can be achieved in a single integrated system.

The project has the following objectives:

1. To create a hybrid deep learning model to classify brain MRI of multiple diseases. The system should be able to recognize and diagnose several neurological disorders

like brain tumors, Alzheimer and multiple sclerosis using images of MRI. This aim is to facilitate the detection of multi-classes of disease in an integrated system as opposed to reducing the system to single disease classification mechanisms.

2. To apply region-of-interest (ROI) extraction, based on segmentation. Segmentation models (U-Net and U-Net++) should be used to detect and isolate critical areas in MRI images using the system. This goal will guarantee that background noise is minimized, only areas that are relevant to the disease are further processed and the classification is more accurate.

3. To carry out deep feature extraction with state-of-the-art CNN designs. The system should be based on deep learning models with a strong capacity like EfficientNetV2-S and dense net 201 to extract high-level features of MRI images. The aim is to learn the intricate patterns and variations of structures in brain images to better represent them.

4. To combine machine learning classifiers in order to achieve high prediction. Algorithms like XGBoost and LightGBM should be used in classifying the extracted features. The goal is to enhance the combination of deep learning and boosting methods to enhance the generalization process, and boost the accuracy of classification.

5. Compare various hybrid models and determine the architecture with the best performance. The project will entail assessing various combinations of hybrids to come up with the most effective model in terms of performance measures. This goal will guarantee the optimum system in terms of accuracy, strength and efficiency.

6. To make visual explanations based on Grad-CAM to achieve interpretability. The system must be able to identify key areas in MRI images which affect the predictions of the model. Such an objective enhances transparency and confidence to the system since the decision-making process can be comprehended by medical professionals.

7. To come up with scalable and trustworthy medical diagnostic model. The overall aim of the project is to develop a system that can be further expanded to real world clinical applications such as early disease detection, decision support and integration with the healthcare systems.

1.3.3 Outcomes

The Project will have the following outcomes.

The results of the project are the indicators of successful implementation of the proposed objectives both technically and practically.

The initial significant finding is the creation of a powerful multi-disease brain MRI classification model by hybrid deep learning methods. The system can accurately detect various neurological conditions based on MRI images, showing that it can be effective in multicuriosity tasks of classifying images into different classes.

The second significant success is the successful applicability of the segmentation-based extractions of ROI. The U-Net and U-Net++ are used to isolate disease-relevant regions and hence enhances the quality of the features and minimizes the influence of the irrelevant background information, thus increasing the classification performance.

Another consequence is that deep learning models can be effectively used to extract features. EfficientNetV2-S and DenseNet201 are examples of architectures that are effective in capturing complex patterns and structural variations in MRI images and hence, the model can learn meaningful representation.

The other important result is the incorporation of machine learning classifiers like XGBoost and LightGBM to make the ultimate prediction. Such hybrid system improves the generalization capability of the system and increases the overall accuracy of prediction, as compared to single models.

The classification accuracy of the project was also high (91% with the most successful hybrid model U-Net++ + EfficientNetV2-S + XGBoost). This shows how the proposed method can be effective in enhancing the work of diagnostics in the classification of multi-diseases.

One more result is the adoption of Grad-CAM representation as a model interpretability. The results are made more transparent and reliable to clinical use by the system where crucial areas in MRI images are pointed out and used to make predictions.

On the practical side, the outcomes of the project will be the full end-to-end framework comprising of segmentation, feature extraction, classification, and visualization. This renders the system more holistic as compared to the conventional methods which only concentrate on classification.

Altogether, the final project result is a smart hybrid system that integrates the ability to detect multiple diseases, analyze by regions, learn deep features, and provide explainable AI in its approach. This makes the work an important contribution towards state-of-the-art medical image processing and contributes to the creation of trustworthy clinical decision support systems.

1.4 Related Applications

1. The proposed multi-disease MRI classification of the brain is highly applicable in the contemporary healthcare, especially in the process of improving accuracy, efficiency, and reliability of diagnosis of a neurological condition. One of the most important spheres of application of this system is clinical decision support systems which can be applied to assist radiologists and neurologists by automating MRI scan analysis, and even diagnostic prediction. The system can assist in reducing the reliance on manual interpretation and variability of the diagnosis depending on human factors by identifying critical areas of the brain with the aid of the visualisation techniques and minimising the variability of the diagnosis depending on fatigue and level of experience. This will ultimately lead to the more standardized and dependable clinical decision making.

2. The second advantage to the use of the system is in screening programs and early detection of diseases. Neurological diseases (e.g., Alzheimer disease, multiple sclerosis) are normally expressed in the subtle change of the brain tissue that can barely be detected through the usual procedure. It will be possible to analyze MRI images automatically and correctly with the help of the proposed system, which will help to define the patterns of the disease in its early stages, when it is impossible to notice significant symptoms. It is a sensitivity to early diagnosis that is extremely important to initiate early intervention to cure the disease, retard its progression and better patient outcome in the long term.

3. The system can also be applied in diagnosis and treatment planning of the brain tumor. Proper and accurate data regarding tumor area, position and development is necessary at the localization and delimitation of tumor areas. This data can help health care providers to plan surgical operations, radiation therapy and chemotherapy. The system provides complete details about the tumor characteristics and facilitates custom-made treatment strategies and the overall effectiveness of health care.

4.Besides, the framework proposed could be applied to the multi-disease diagnosis system at the hospitals and the diagnostic stations as well. This model can support various neurological conditions in a single architecture as compared to traditional systems, which are only designed to support particular diseases. This will minimize the use of individual diagnostic tools, enhance the workflow, and allow the analysis of brain MRI scans comprehensively within one system. The latter kind of integration can be particularly helpful in a hectic clinical environment where the quick and precise diagnosis is vital.

5. It can also be used in remote health care and telemedicine as well. With the need to adopt digital healthcare solutions, patients will have access to posting MRI scans to online systems, and obtain an automatic diagnosis response without necessarily visiting hospitals. It can be particularly important to under-serviced and rural communities where special medical specialists are not easily accessible. The system promotes remote diagnosis which consequently results in more people having access to health care and the geographical obstacles are reduced.

6. In addition, the suggested system can be effectively utilized in medical research and data-driven healthcare. The framework will help the researchers to analyze the large amounts of MRI data and identify the trends concerning different neurological disorders. It may lead to a better understanding of the disease progression, emergence of biomarkers, as well as the emergence of novel diagnostic approaches. The combination of hybrid models also opens possibilities to experiment with more efficient combinations of deep learning and machine learning models in medical imaging.

7. It can also be used in explainable artificial intelligence (XAI) in healthcare. Use of visualization such as Grad-CAM would make the system provide interpretable results, by displaying the regions that cause the predictions. Such openness is a key to establishing trust between medical workers and making AI-based systems become a

part of clinical practice. It also helps in checking model choices, and establishing the potential errors.

8. The system can also be used as a platform of education and training to medical students, and health care practitioners. It assists students to study the complicated neurological illnesses and enhance diagnostic abilities visualising the localities involved by the illness and giving them the outcomes of the classification. This will make the system useful in the diagnosis and as well as academic and training.

9. Lastly, the future automated healthcare systems can be based on the proposed framework. It can be applied to real-time applications, connect with hospital information systems and other diagnostic modalities like CT scans and clinical data. This scalability has made it a promising solution to develop intelligent healthcare ecosystems that can be applied to efficiently detect, monitor, and manage diseases.

CHAPTER-2

Review of Literature

2.1 Description of Existing Systems

Hybrid machine learning (Hybrid) has been suggested to detect brain tumors in MRI images, in which segmentation characteristics are obtained with both 2D and 3D architectures of U-Net. The paper indicates that segmentation is important and is capable of isolating tumor regions and enhancing the quality of features, as explained in [1]. This is then classified into advanced texture and statistical characteristics. K-Nearest Neighbor (KNN) and Gradient Boosting Classifier (GBC) are two algorithms that are ensemble-based, meaning that the authors use them together to improve prediction accuracy. As the model demonstrates the better performance, it can be applied only to detect tumors, and it does not deal with the multi-disease classification.

A convolutional neural network-based deep learning method to classify brain tumors has been proposed, which automatically learns spatial characteristics of MRI images. In [2], the usefulness of CNNs in deriving hierarchical image features is highlighted, and there is no necessity to engineer features manually. The model however, operates on whole images without concentration to the region of interest which brings in background noise. This drawback can decrease the accuracy of the classification, particularly with complicated medical images. Also, the methodology is limited to the classification of single diseases.

To enhance the performance of brain tumor detection, a hybrid method that involves CNN-based feature extraction and conventional machine learning classifiers have been proposed. It is discussed in [3] and seeks to combine the advantages of both methods in an integration of the deep features with efficient classifiers. This approach enhances the classification accuracy as opposed to individual CNN models. Nonetheless, the system continues to rely on rudimentary preprocessing algorithms and does not have sophisticated segmentation algorithms. Consequently, it might not be very effective in dealing with intricate tumor variations.

To identify different kinds of brain diseases with deep learning models, a multi-class classification system is suggested. It is shown in [4] that it is feasible to deal with a variety of neurological conditions using a single framework. Nevertheless, the model is problematic because of the scarcity of the diversity of the datasets and inappropriate optimization of features. The lack of region-based learning also decreases its capacity to differentiate similar classes of diseases. This demonstrates the necessity to have more sophisticated hybrid systems.

U-Net with segmentation-based approach has been applied to detect tumor sections of MRI images prior to classification. These issues are discussed in [5] where the significance of region-of-interest extraction to enhance accuracy is explained. The model eliminates noise and improves the quality of features by employing the relevant regions. But the system is tailored to detecting tumors and not to other neurological disorders. It is also not integrated with the advanced classification techniques.

Deep convolutional neural networks have been used to classify medical images to automatically identify useful features in MRI data. This ability of CNNs to acquire complicated patterns is emphasized in [6] and is more efficient than the ordinary approaches. Although these are the strengths of the model, it is disadvantaged by over fitting because of the limited training data. It also fails to implement hybrid learning methods, thus constraining its generalizability to new datasets.

A hybrid deep learning model has been suggested to classify Alzheimer disease with a variety of pre-trained models like EfficientNet, Xception, and MobileNet. Ensemble learning can be used to enhance classification accuracy, a fact that is proved in [7]. The model has the capacity of detecting fine trends in neuroimaging data, which is crucial in detection at an early stage. The system, however, lays emphasis on Alzheimer disease only, other neurological disorders are not taken into consideration.

Neuroimaging-based classification algorithms have been investigated to identify the Alzheimer disease, based on the MRI data, by evaluating structural changes in the brain. In [8], the importance of MRI in determining brain atrophy and degeneration is highlighted. Although the model would offer valuable information on how the disease advances, it is not very effective at detecting the disease at an early stage because of

the minor variations. It is restricted by the predefined features, which reduce its flexibility in dealing with complex data.

The machine learning-based system is developed on the handcrafted MRI features comprising of texture and statistical properties to classify diseases. In [9], they used traditional feature extraction methods, which are computationally efficient. Nonetheless, these characteristics are not adequate to identify intricate structures in medical images. Lack of deep learning reduces the capabilities of the model to be accurate.

MRI images and conventional machine learning algorithms have been used to create a classification system of multiple sclerosis. This is discussed in [10] where the variability and detection of MS lesions are discussed. Although the system has moderate accuracy, it has difficulty in the differences in lesion sizes, shapes and intensities. Its inability to extract deep features is a weakness that prevents its application in complicated situations.

A segmentation model that is a deep learning has been proposed to detect brain lesions in MRI images. In [11], the significance of proper segmentation in medical diagnosis is emphasized. The model is also able to isolate disease affected areas, which enhances disease pattern. Nonetheless, it lacks a classification aspect thus complete end-to-end diagnosis is missing.

DenseNet has been used in medical image classification to enhance the reuse of features and gradient flow. The merits of thick connectivity are reported in [12], which results in enhanced performance over the conventional CNNs. The model however is very computationally intensive and lacks segmentation. This restricts its capacity to target the areas of relevance.

Medical image analysis has been done using EfficientNet because of its efficiency and scalability. The model is more accurate using less parameters as described in [13]. Nevertheless, it is mostly about classification and lacks segmentation and hybrid learning methods. This diminishes its capacity of capturing features specific to regions.

Explainable AI methods like Grad-CAM have been investigated to visualise deep learning model predictions in medical imaging. The need of interpretability in clinical applications is addressed in [14]. The approach identifies key areas that contribute to the predictions, which enhances confidence in AI systems. Nonetheless, it is generally used on single-model and does not consider hybrid models.

A hybrid solution integrating both deep learning and machine learning has been suggested to do the disease classification. The advantages of implementing the combination of methods are described in [15], enhancing the overall performance. Nonetheless, the system does not have an end-to-end pipeline (encompassing segmentation, feature extraction and explainability). Neither does it discuss the classification of multi-diseases within a single framework.

Early works were devoted to the development of other imaging approaches that could be used to detect abnormalities in the brain. A combined technique for the imaging of brain injuries based on both qualitative and quantitative approaches has been suggested[16]. It allows one to increase the speed and accuracy of abnormality identification; however, such techniques imply solving complicated inverse problems .

Deep learning is now gaining increasing popularity. As such, researchers have started to devote much attention to the development of automatic diagnosis techniques for diseases using magnetic resonance imaging data[17]. In particular, a hybrid approach based on transformers and recurrent neural networks has been devised to provide highly accurate classification results; besides, the suggested framework utilizes techniques from explainable AI.

Moreover, a hybrid deep learning algorithm has shown great potential in brain tumor detection. The hybrid deep learning algorithm that uses a convolutional neural network (CNN), in combination with multiple machine learning algorithms, has yielded better results than conventional methods in terms of tumor classification accuracy, precision, and recall [18]. Nevertheless, feature and classifier selection can be significant challenges for such systems.

More recently, researchers have explored hybrid models by including segmentation, feature extraction, and optimization. For instance, a hybrid attentive architecture that

includes preprocessing, segmentation, transformer feature extraction, and deep learning classification has yielded excellent results in multiclass tumor classification tasks[19]. Moreover, researchers have used explainable AI techniques to enhance model interpretability and clinical confidence .

Likewise, hybrid ensemble deep learning models have been developed for the classification of Alzheimer's disease using multiple architectures. Hybrid ensemble deep learning models have achieved outstanding accuracy and employed Grad-CAM to focus on critical brain areas [20].

2.2 Summary of Literature Study

The analysis of the existing literature reveals substantial advancement in the medical image analysis field for detecting diseases in neurology through machine learning and deep learning applications. Several studies have established that convolutional neural networks (CNNs) and hybrid models are more efficient in extracting valuable features from MRI scans to classify and diagnose specific neurological diseases, as shown in [1], [2], and [3]. The segmentation technique through Unet models has revealed that the isolation of ROI improves the precision of diagnosis since background noises are minimized [4], [5], [6].

Some of the researches on the topic have employed hybrid models by using deep learning and machine learning ensembles to predict neurological diseases. The models have proven to be more effective than the independent models because of the integration of different algorithms' strengths [7], [8], [9]. The boosting method, including LightGBM and gradient boosting, is highly successful in dealing with deep learning models' high-dimensional complex features.

A review of the existing literature has highlighted that very few comprehensive comparative studies exist that attempt to analyze the efficiency of different hybrid architectures at identifying the best way to treat patients using those techniques. Some studies demonstrate that using an individual technique (for example, EfficientNet, DenseNet, or various forms of ensemble learning) results in improvements in patient

care; however, all of these approaches lack a common framework for integrating and utilizing the common elements of each approach [10],[11].

As a result, there is an opportunity for the researchers to develop an advanced hybrid architecture that is able to effectively communicate the results of segmentation, deep learning-based feature extraction, and, finally, the machine learning-based classification of patients with several different neurological conditions using a common framework. A unified framework approach should allow for improved classification accuracy by using region-based learning (ROIs) and should provide the clinical end-user with the ability to interpret those results. In this context, the proposed research will help fill this void by creating an effective classification model of multiple neurological conditions by using hybrid architecture methodologies, through the application of explainable AI methods [12],[13]

The literature suggests a lack of thorough analyses comparing hybrid architectures, which creates challenges and limits identification of hybrid methods that deliver optimal outcomes. Although each individual study demonstrates an increased rate of success using a specific approach such as EfficientNet, DenseNet, or ensemble learning; none of them have established a complete and seamless way to integrate all parts of each approach [14][15].

It was concluded that there is a need for a more complex hybrid solution that integrates segmentation with deep learning-based feature extraction as well as machine learning classification within a single framework for several neurologic disorders to provide measurable improvements in accuracy due to ROI-based learning algorithms and to produce meaningful results for use by clinicians. The proposed project will create a multi-disease classification model using hybrid methods as well as explainable AI [17], [19].

CHAPTER-3

Proposed Method

3.1 Design Methodology

The given system adheres to a grandiose and systematic approach to the creation of a sophisticated multi-disease brain MRI classification system, comprising of several steps of data processing, deep learning, and machine learning. It starts with the gathering of MRI data that is related to various neurologic conditions such as brain tumors, Alzheimer disease, and multiple sclerosis across multiple publicly available sources. These datasets are subsequently pooled into one dataset so that the model can learn various patterns that are related to different diseases. Medical datasets are typically imbalanced in terms of classes (with some classes represented less frequently than others) so a balancing approach is used to make sure that each class has the same number of images. This assists in minimizing bias in training and makes sure that all classes are learned in the model. The data is further segmented into training, validation and test datasets to enable the model to be learned, tuned and evaluated.

The next step after data preparation is preprocessing to standardize the MRI images and enhance the input data. This involves scaling of all images to a fixed resolution, equalization of pixel intensity values and conversion of images to a uniform format fit to input the model. These preprocessing measures are necessary to minimize the variability of the data and stabilize convergence when training the model. After preprocessing, the system applies the segmentation process based on the advanced architectures like U-Net and U-Net++, which are dedicated to the analysis of medical images. Segmentation is essential in isolating the region of interest, e.g. tumors or lesions, by segregating relevant structures and the background. This will increase the concentration of the model on disease-specific areas and reduce the impact of any irrelevant information.

The (segmented) output is then used to refine the original MRI images using the predicted masks effectively only important regions are retained and the background noise is eliminated. This area oriented method enhances the quality of features obtained

in later steps. Refined images are then subjected to deep learning architectures (EfficientNetV2-S and DenseNet201) that can extract high-level and hierarchical features of complex medical images. These architectures have been developed in order to recreate complex spatial patterns, textures and structural variations in MRI scans, crucial in providing precise classification of the disease. The feature extraction procedure converts raw image data to a succinct and informative representation which can be efficiently used by machine learning classifiers.

The feature vectors obtained are then classified through the advanced machine learning algorithms like XGBoost and LightGBM. These algorithms are founded on boosting methods, where a combination of weak learners is used to create a strong predictive model. The method boosts the accuracy of classification, generalization, and minimizes the chance of overfitting. Deep learning to extract features and machine learning to classify the features constitute a hybrid network that uses the merits of both methods. The training phase involves the use of different optimization algorithms, including the adaptive learning rate scheduling, early stopping, and data augmentation, to enhance the performance and stability of the models. These methods allow reaching a higher convergence rate and avoid the model to memorize the training data.

System performance is measured with the help of several evaluation measures including accuracy, precision, recall, F1-score, as well as confusion matrix that allow a holistic analysis of the classification performance in various types of diseases. Besides classification, the system also includes visualization features like Grad-CAM, to offer interpretability of the model showing the areas of MRI images that contribute to the prediction made by the model. This is especially significant in medical use, where one needs to have a chance to comprehend the rationale behind predictions to earn the trust of healthcare professionals. The system is the last stage where all of the components are put together into a single pipeline to process input MRI images and segment, extract features, classify diseases, and produce interpretable outputs. This end-to-end design guarantees that the system is not just correct but also reliable, scalable and appropriate in real world clinical use.

The model with more accuracy and best performance will be used to build the Grad-CAM and the whole system will be deployed in a web application. The web application

will be developed by using the Flask for backend and HTML, CSS, JavaScript for frontend**.**

3.2 System Architecture Diagram

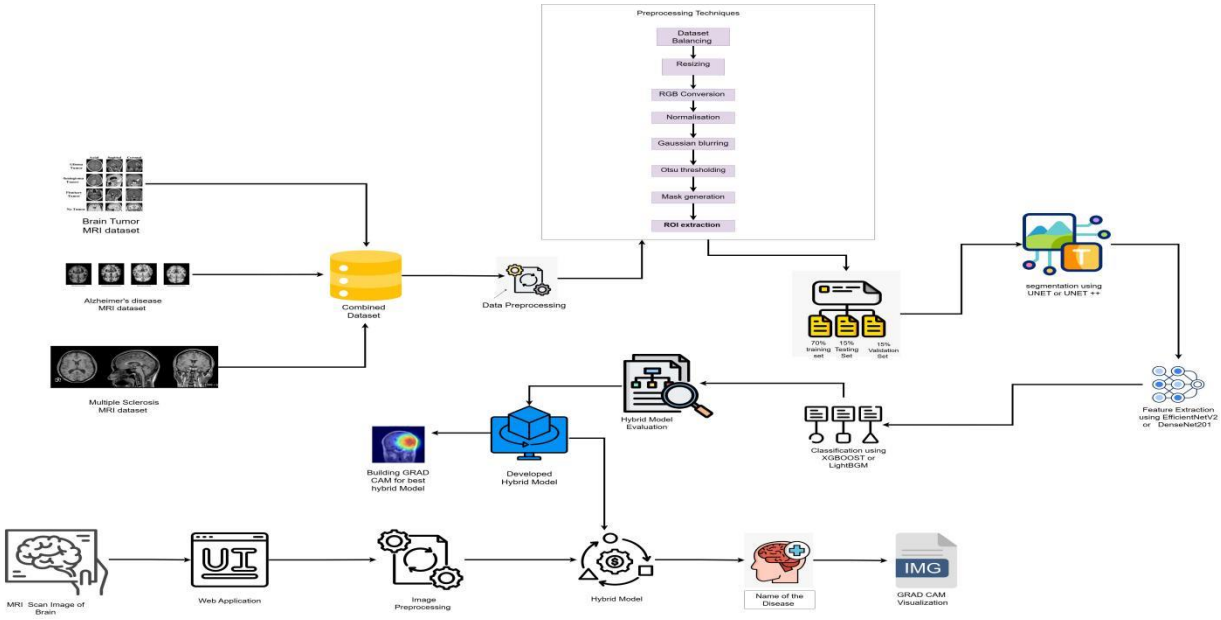

Figure 3.1: System Architecture Diagram

Figure 3.1 shows the general layout of the proposed system, which is a hybrid, multi-stage pipeline that can classify brain MRI images with high accuracy and interpretability as having multiple diseases. The architecture combines various elements such as integrating datasets, preprocessing, segmentation, deep feature extraction, machine learning-based classification, model evaluation, and visualization. This system is designed in a chronological but modular fashion where each stage can play on its own towards the overall performance and still be flexible to add improvements in the future.

The architecture commences by acquiring and incorporating various MRI images of brain tumors, Alzheimer disease and multiple sclerosis. Such datasets are pooled together to make up a single dataset to make sure that the model is subjected to a wide variety of disease patterns and imaging variants. The dataset balancing strategy is used to have the same number of samples in each of the classes, which is necessary to avoid bias and to achieve fair learning in the process of training. This step is the basis of the

system because the quality of data and its diversity is the determining factor of the success of the following stages.

After preparing the data, it is integrated with a complete preprocessing pipeline that normalizes and improves the input data. Preprocessing stage comprises of resizing images to a fixed size, transforming into a uniform color space, equalizing the intensity values of pixels, and noise reduction methods. We also apply thresholding and morphology to create pseudo-masks that are useful in determining the important areas in the MRI images. These preprocessing operations are essential to minimize variability, enhance signal-to-noise ratio, and make sure the input data is compatible with deep learning models.

The second significant part of the architecture is the segmentation module that is very important in the extraction of the region of interest (ROI) in MRI images. This is done using advanced architectures like U-Net and U-Net++, which are capable of localizing fine-grained spatial representations and boundary details. The segmentation model produces binary masks that identify the disease-affected areas like tumours or lesions. Such masks are then overlaid on the original images to generate refined inputs that only have the most relevant information. This ROI-based methodology is highly effective in enhancing classification accuracy due to the removal of background noise and orientation of the model towards disease-specific characteristics.

Following the segmentation, the architecture turns to the feature extracting step, during which deep learning models like EfficientNetV2-S and DenseNet201 are used. These are hierarchical feature representations that are learned on images, both at the low-level e.g. edges and textures, and the high-level e.g. shapes and structural patterns. EfficientNetV2-S offers the trade-off between accuracy and computational efficiency by scaling efficiencies, whereas DenseNet201 enables improved feature propagation and reuse via dense connectivity. The results of this step are a collection of high-dimensional feature vectors that represent significant data on the input MRI images.

The system does not use the deep learning models to carry out classification directly but instead uses a hybrid method to incorporate machine learning classifiers like XGBoost and LightGBM. These algorithms act on the extracted feature vectors and make multi-class classification, learning some complex decision boundaries. The

techniques of boosting are especially useful when dealing with high dimensional data and to enhance the accuracy of prediction by assembling a number of weak learners into a strong ensemble. The hybrid approach improves the generalization and strength of the system over individual deep learning models.

There is also a specific model evaluation portion of the architecture, where the performance of various hybrid models is evaluated through the use of metrics like accuracy, precision, recall, F1-score and confusion matrix. This process of evaluation can be used to comparatively analyze different combinations of models, and the best-performing architecture can be identified. A decision on the best model will be made based on how well the model can perform accuracy and be consistent across the classes.

A key aspect of the suggested architecture is that it has explainable AI in the form of Grad-CAM visualization. After choosing the model with the highest performance, the Grad-CAM is used to produce heatmaps indicating the areas in the MRI images most important to the predictions given by the model. This increases the system interpretability and generates quality information to medical practitioners so that they can be able to testify and trust the decisions of the model.

Lastly, the system itself is seamlessly connected into a single end-to-end pipeline, which takes input MRI images and performs all processing steps, including preprocessing and segmentation up to classification and visualization. The architecture is designed to be scaled and thus it is possible to add new components or enhancements and this will not impact on the overall workflow.

On the whole, the offered system architecture is an efficient method of integrating various sophisticated methods into a unified system, which will overcome the shortcomings of the current systems and offer an effective approach to classifying multi-disease brain MRI.

3.3Description of Algorithms

3.3.1 EfficientNetB7

The algorithm in Mini Project1 is grounded on a deep learning architecture with the EfficientNetB7 architecture to categorize brain tumor and Alzheimer disease MRI

images into multiple classes. The methodology is based on a systematic pipeline that involves the balancing of dataset, its preprocessing, deep feature extraction and classification. The main goal of the algorithm is to learn intricate spatial and structural features that occur in MRI images and project them onto disease categories with a highly optimized convolutional neural network.

Algorithm:

Input: The data that will be inputted into the proposed system will be brain MRI images of various datasets depicting various neurological conditions namely brain tumor and Alzheimer disease types. These images are of eight different classes and they are combined to form a single dataset so as to be able to classify them into multiple classes. Given the fact that real-world medical datasets are usually subject to class imbalances, data augmentation methods are used to make sure that samples are evenly distributed among all classes. Such methods involve geometric and spatial manipulations like rotation, translation, zooming, and flipping, which assist the model to acquire the features of being invariant to various orientations and circumstances.

The stratified sampling is again used to split the dataset into training, validation, and testing subsets to maintain the class distribution of all the splits. Preprocessing operations are carried out before feeding the data into the model, such as resizing images to a common dimension, and normalizing pixel intensity values. The importance of normalization as shown in Equation (1) lies in ensuring numerical stability and more rapid convergence during training because it scales pixel values to a common range. This phase is to guarantee that the input data is normalized, balanced, and capable of being extracted into features with optimal deep learning.

$$X_{norm} = \frac{X}{255} \tag{1}$$

Method: It is anchored on a deep convolutional neural network framework based on EfficientNetB7 that is meant to ensure a high accuracy but has computational efficiency with compound scaling. EfficientNet does not rely on independent scaling of depth, width, and resolution as in the traditional CNN architecture, but rather balances the scaling.

This enables the network to utilize computational resources efficiently and enhances the feature representation. The input MRI images are fed through a series of convolutional layers comprising of filters which extract features in the form of edges, textures, and structural patterns. Low-level features like intensity variations and boundaries are captured by the early layers whereas high-level semantic features like tumor regions and brain abnormalities are captured by deeper layers.

Mathematically, the feature extraction process can be represented as shown in Equation (2):

$$F = f(X_{norm}) \tag{2}$$

Where:

F is the feature maps extracted

f represents the transformation carried out by the EfficientNetB7 model.

These are multi-dimensional representations that have spatial and channel wise data of the input image.

The resulting feature maps are flattened into a one dimensional vector as represented in Equation (3):

$$f_{flat} = flatten(f) \tag{3}$$

This step transforms spatial information into a form that can be used in fully connected layers that allow the model to do classification. The resulting, flattened feature vector is further run through dense layers, in which linear transformations are used to project features into class-specific representations, as shown in Equation (4):

$$\mathrm{Z} = \mathrm{W}.f_{flat} + \mathrm{b} \tag{4}$$

The Rectified Linear Unit (ReLU) activation function is employed to add non-linearity to the model and to allow the model to learn intricate decision boundaries as represented in Equation (5):

$$\mathrm{ReLU(x)} = \max(0, \mathrm{x}) \tag{5}$$

The concept of regularization is substantial to deep learning models, particularly when the large architectures (e.g., EfficientNetB7) are involved. In order to avoid overfitting,

dropout is used, in which a random proportion of neurons are dropped out during training as described in Equation (6):

$$F' = F * r \tag{6}$$

Where r is distributed with a Bernoulli. This is to make sure that the model is not dependent on particular neurons and enhances generalization.

The last classification layer applies the Softmax function to transform the output logits into probability distributions over all classes as represented in Equation (7):

$$P_i = \frac{e^{z_i}}{\sum_{j=1}^{12} e^{z_j}} \tag{7}$$

This makes the output values to be normalized and interpretable as probabilities. The categorical cross-entropy loss function is used as a guide in the training process as it quantifies the difference between the predicted and actual labels as shown in Equation (8):

$$L = -\sum_{i=1}^{8} y_{i \log(\widehat{y_i})} \tag{8}$$

The Adamax optimizer, an extension of the Adam optimizer, is employed to optimize the model parameters, and is especially useful when the parameter space is high-dimensional as represented in Equation (9):

$$\theta = \theta - \eta \cdot \nabla L \tag{9}$$

Also, there is a learning rate scheduling strategy that gradually reduces the learning rate during epochs as described in Equation (10):

$$n_t = n_0 \cdot (0.95)^t \tag{10}$$

This will aid in the attainment of stable convergence and avoid oscillations during training. The general approach is a synthesis of efficient feature extraction, non-linear transformations, regularization, and optimization strategies to construct a solid classification model.

Output: The system gives a predicted class label as output which is the most likely class of the input MRI image which is one of the eight categories of brain tumor or Alzheimer disease. Not just does this output yield classification results but also indicates the degree of the model in its prediction, which can be used to aid in medical diagnosis and

decision-making processes. It will predict the class label by using the softmax function where this function will return the class with higher probability.

3.3.2. EfficientNetB3-Based Classification

EfficientNet-B3 is the preferred main classifier due to its accuracy and efficiency in computation. It belongs to the EfficientNet family, that simultaneously scales model depth, width, and resolution via a compound scaling technique. This enables EfficientNet-B3 to be highly accurate and does not require a large number of parameters to be added.

Input: The data to be used in the EfficientNetB3 based system includes brain MRI scans of various datasets of brain tumors, Alzheimer disease and multiple sclerosis. These data sets are merged to form one dataset with 12 different classes to allow the model to learn a broad variety of neurological disease patterns. To achieve a balanced learning, every class will be modified to have equal number of images by using sampling methods.

Stratified sampling divides the dataset into training, validation and testing subsets to maintain the class distribution. The size of all images is adjusted to a constant size of 244 x 244 pixels, which works with EfficientNet-based architectures. A normalization function as shown in Equation (11) is used to preprocess using the standard values of pixel intensities:

$$X_{norm} = \frac{(X-\mu)}{\sigma} \tag{11}$$

This normalization enhances numerical stability and guarantees effective gradient propagation when training.

Method: EfficientNet uses scaling of compounds as defined in Equation (12) to scale uniformly network depth, width and input resolution as given in the following where a single scaling coefficient is used to scale the overall model size without compromising on the best accuracy-efficiency trade-offs.

$$\text{Depth}=\alpha^{\phi},\ \text{Width}=\beta^{\phi},\ \text{Resolution} = \gamma^{\phi} \tag{12}$$

This scaling approach enables the network to make an extraction of rich and meaningful features of MRI images but with computational efficiency. The input images are fed

through several convolutional layers which learn hierarchical representations. The low-level features that are captured by the early layers as shown in Equation(13) include edges, gradients and textures whereas the high-level features that are captured by the deeper layers include tumor regions, lesion structures and abnormal brain patterns.

$$F = f(X_{norm}) \tag{13}$$

The feature maps thus obtained are shrunk by global pooling operations that transform spatial feature maps into small feature vectors with help of Equation (14):

$$F_{pool} = GlobalPooling(F) \tag{14}$$

A batch normalization method is used to stabilize training, and minimize internal covariate shift by using the Equation (15) :

$$X' = \frac{X-\mu}{\sqrt{\sigma^2+\epsilon}} \tag{15}$$

The extracted features are then mapped into a class-specific representation by fully connected layers as represented in Equation (16):

$$Z=W\cdot F_{pool}+b \tag{16}$$

The non-linearity is added by the ReLU activation function as expressed in Equation (17):

$$ReLU(x)=max(0,x) \tag{17}$$

To avoid overfitting, dropout regularization is applied with the help of Equation (18) When training, the neurons are randomly disabled:

$$F'=F\cdot r \tag{18}$$

The last classification layer employs the Softmax in Equation (19) to generate probabilities over the 12 possible classes:

$$P_i = \frac{e^{z_i}}{\sum_{j=1}^{12} e^{z_j}} \tag{19}$$

Categorical cross-entropy as expressed in Equation (20) loss is used to train the model:

$$L = -\sum_{i=1}^{12} y_{i \log(\widehat{y_i})} \tag{20}$$

Adaptive optimization techniques are used to optimize the model parameters by expressing the Equation (21):

$$\theta=\theta-\eta\cdot\nabla L \tag{21}$$

Additional performance and convergence benefits of fine-tuning and learning rate scheduling are also provided.

Output: The EfficientNetB3 model predicts the output of class label which is one of the 12 brain disease categories. This model compiles a probability distribution of all the classes and the most probable one is used as the final prediction. This allows proper multi-classification of neurological diseases based on MRI scans.

3.3.3. AlexNet-Based Classification

The deep Convolutional Neural Network design known as "AlexNet" made a very big impact on the success of computer vision projects. It proved that deep learning methods could solve difficult problems in the area of classification. AlexNet contains several convolution layers, pooling layers, and entirely connected layers that allow for learning features from the input image hierarchy or layers into a model so that the trained model can process an image in its entirety rather than a small portion of it. It makes use of "ReLU" activation functions to add non-linearity to the model, which enables faster convergence to solution (training time) and a better representation of features from input. Training of AlexNet is also aided by regularization methods like dropout, which prevent the model from overfitting, resulting in enhanced generalization of the characteristics of the input data. The combination of the design of AlexNet and its ability to learn/elucidate low/high-level features have made this primary architecture relevant within computer vision.

Input : Preprocessed brain MRI images of combined data of brain tumor, Alzheimer diseases, and multiple sclerosis form the input to the system based on the AlexNet. The data is balanced so that there is equal representation of all 12 classes. The images are all downsized to a standard size of 227 x 227 pixels, the standard size of the input layer used in the AlexNet architecture.

Scaling of input values is done by pixel normalization. This guarantees uniformity and efficiency in training.

Method: The AlexNet is a classical deep convolutional neural network with several convolutional, pooling and fully connected layers. It is created to acquire hierarchical feature representations of images by a series of transformations.

The convolution operator was defined in Equation (22):

$$F(i,j) = \sum_m \sum_n X(i+m, j+n) \cdot K(m,n) \tag{22}$$

This operation makes the network to derive spatial features like edges, textures, and intensity variations. The more the depth of the network, the finer details of a pattern it captures including shapes and structural abnormalities within brain MRI images.

ReLU activation is a non-linear activation and used to introduce the non linearity in the model during the training of the model, as represented in Equation (23).

$$\mathrm{ReLU}(\mathbf{x}) = \max(0, \mathrm{x}) \tag{23}$$

Pooling layers downsample the spatial dimensions and keep large features, as shown in Equation (24):

$$P\ (i,j) = \max_{(m,n)\in\Omega} F(i+m, j+n) \tag{24}$$

This enhances the computational efficiency and offers translation invariance. The feature vectors are flattened into a one dimensional vector, as represented in Equation (25):

$$f_{flat} = flatten(f) \tag{25}$$

High-level feature interpretations are done by fully connected layers, as shown in Equation (26):

$$\mathrm{Z} = \mathrm{W}.f_{flat} + \mathrm{b} \tag{26}$$

The use of dropout helps to avoid overfitting by dropping neurons at random with training.

The last classification is done through the use of softmax of 12 classes as computed according to Equation (27):

$$P_i = \frac{e^{z_i}}{\sum_{j=1}^{12} e^{z_j}} \tag{27}$$

The categorical cross-entropy loss is used to train the model as described in Equation (28):

$$L = -\sum_{i=1}^{12} y_{i \log(\widehat{y_i})} \tag{28}$$

Learning (gradient-based) is done to optimize as derived in Equation (29):

$$\theta=\theta-\eta\cdot\nabla L \tag{29}$$

Output: The AlexNet model output is a class label prediction of one of the 12 brain disease classes. The model creates probability scores of the different classes and the highest probability is taken as the final prediction. This allows categorizing of the neurological diseases according to acquired visual patterns through MRI images.

3.3.4. LIME(Explainable AI Model)

Local Interpretable Model-Agnostic Explanations (LIME) is used as an interpretability method to increase the transparency, reliability and trust of the proposed brain MRI classification system. High-accuracy is not the only requirement in medical image analysis, it is also important to comprehend and explain the choices taken by the model. Deep learning models are frequently referred to as a black-box system since the reasoning that takes place in them cannot be readily interpreted. LIME can alleviate this shortcoming by giving explicit and user-friendly explanations of each prediction without making any alterations to the underlying model.

LIME is a model-independent approach, i.e. it can be used with any machine learning or deep learning model regardless of the architecture. Rather than describing the whole model on an international scale, LIME tries to interpret the predictions on a local scale i.e. around a given input image. This localized explanation is most applicable in medical fields, where clinicians require knowing the rationale of a particular diagnosis, and not the general behavior of the model.

The essence of LIME is to estimate the behaviour of a complicated model with a simpler and interpretable model around a particular input. In the case of image data, LIME breaks down the MRI image into various meaningful components referred to as superpixels. These superpixels are clumps of pixels that have similarities and it is easier to explain them than a single pixel does.

Working of LIME

The use of LIME has a number of steps to follow in order to interpret model predictions:

Input Image Selection: One MRI image is picked and the model trained provides a prediction to the image. This is the prediction that is used as the reference to explain.

Superpixel Segmentation: The input image is segmented into several superpixels and a superpixel is an area of similar texture or intensity. The segmentation assists to analyze the significance of various regions in an organized way.Perturbed Samples are generated in the same manner as with ordinary samples.

LIME produces multiple variants of the original image as the result of random masking or editing of various superpixels. The variations are slightly modified versions of the input image, and the sensitivity of the model to different parts of the input image can be studied.

Predicting on Perturbations: All perturbed images are run through the trained model to see how the prediction varies. In case the removal of a certain area has a significant influence on the prediction, then that area is deemed as important.

Local Interpretable Model Building: Using the generated samples and their respective predictions, LIME constructs an easy to understand model that is an approximation of the complex model in the local area. The model gives a score of importance to each superpixel by the contribution given to the prediction as expressed in Equation (30).

$$g(z) = \omega_0 + \sum_{i=1}^{n} w_i z_i \tag{30}$$

where:

- $g(z)$= local interpretable model
- z_i= binary indicator for superpixel presence
- w_i= importance weight

Determination of key Regions: The use of the importance scores helps to determine the superpixels that have a positive or negative impact on the prediction. Areas that make more contribution to the classification decision are the areas that are of higher importance, as defined in Equation (31).

$$\text{Importance}(i) \propto wi \tag{31}$$

Visualization of Explanation: Lastly, LIME identifies the strongest superpixels to the original MRI image. These areas of emphasis show the areas the model paid attention to when coming up with its prediction, and this gives it a visual explanation, which is easy to interpret.

LiME : LIME in the case of brain MRI classificationRegions of the brain that are highlighted by LIME are often of clinical importance such as tumors, lesions or abnormal brain tissues. This enables the medical professionals to determine whether the model is attending to the anatomical structures of interest hence boosting confidence in the system.

Altogether, LIME helps to convert the deep learning model into a black-box system to an interpretable and trustworthy framework by offering localized explanations of a specific prediction. It can be of great use in healthcare dynamics as its visual justification of model decisions is essential to adopting and clinical testing its findings.

3.3.5 Hybrid Model 1(UNet++ + EfficientNet V2-S + XGBOOST)

It is a hybrid architecture that integrates segmentation, deep feature extraction and classification using machine learning to classify brain MRI images efficiently and accurately with multiple diseases. The method is to first of all localize the parts of the disease-relevant information by segmentation and then to extract the discriminative features and categorize them using the powerful boosting algorithm. This mixture will enhance the accuracy and generalization since it will reduce the background noise and augment the quality of feature.

Input: The input data that goes into training the hybrid model includes brain MRIs from a variety of sources where the images portray a variety of brain tumors (such as gliomas), as well as images showing other progressive neurological diseases such as Alzheimer’s Disease, and images from patients with Multiple Sclerosis (MS). These datasets were combined to create one dataset that included images belonging to multiple classes and the method of balancing was used to create an equal number of images for each class; this reduced any potential bias and improved the generalization ability of the model across each disease type.

The total dataset was split randomly into training, validation and testing data using

stratified sampling to ensure that all image class splits were equally represented in each of these 3 datasets. Resizing and normalization techniques were performed on the input data to bring them to some standard in preparation for model training. Specifically, the normalization of pixel values was completed to bring them back to a range of suitable values

In addition, segmentation labels that are necessary for guiding the segmentation model were created with pseudo mask generation and thresholding techniques.

Method: Pseudo masks are created by the thresholding techniques such as Otsu thresholding to differentiate foreground and background parts. They are used in cases of weak supervision, where it is not necessary to annotate data manually, although it can also provide valuable information regarding segmentation.

The U-Net++ is trained on the pseudo masks generated and enhances flowing of features with dense skip connections. The model is trained to acquire a mapping between an input image (X) and segmentation mask(M), as formulated in Equation (32).

$$M=f(X) \tag{32}$$

The resultant MRI map is superimposed with the segmentation mask prediction map to obtain the region of interest, as computed using Equation (33):

$$X_{ROI} = X \odot M \tag{33}$$

ImageNet pretrained weights are used to initialize EfficientNetV2-S. This gives the model the ability to leverage existing trained low-level features such as edges and textures, reduces training time and convergence.

The last layers are optimised to fine-tune the network on the MRI images using the differential learning rate. This allows the model to adapt to domain-specific trends but the model has valuable pretrained features.

This eliminates the classification head and the network is used as a feature extractor to generate high level feature vectors. These features capture complex spatial and structural information from MRI images.

The characteristics obtained are fed into the XGBoostclassifier. XGBoost is a sequential tree-based model, which trains numerous decision trees, each tree correcting the error of the previous ensuring an improved classification and reduced overfitting as described in Equation (34).

$$L = \sum_{i=1} l(y_i, \hat{y}_i) + \sum_{k=1}^{k} \Omega(f_{\ k}) \tag{34}$$

where:

- l= loss function
- Ω= regularization term

Class probabilities are computed with the help of the softmax function, as given in Equation (35):

$$P_i = \frac{e^{z_i}}{\sum_{j=1}^{12} e^{z_j}} \tag{35}$$

In which (P_i) is the likelihood of the class (i), so that the results of all the classes are normalized.

Output : A hybrid model predicts a patient's brain disease by producing a class label (one of many categories of brain diseases) for a patient as input of the model. This hybrid model creates a probability distribution among all of its possible classes; therefore, the predicted class with the highest probability is selected as the final predicted class for the patient's MRI image.

This approach also allows interpretation of the prediction of a patient's brain disease through visual methods like Grad-CAM. Grad-CAM uses images of the regions of an MRI image most responsive to the predicted output of the hybrid model to provide visual evidence of why this prediction was made as a way to add to trust and use of the hybrid model in a medical setting by providing additional visual evidence of why a particular class was assigned to an MRI image.

3.3.6 Hybrid Model-2(Unet + DenseNet201 + LightGBM)

This hybrid model consists of the similar basic multi-stage pipeline but with some other segmentation, feature extraction, and classification components. It will be testing and will provide a comparative analysis of the original hybrid structure.

Input : The multiple datasets from which the hybrid model input is derived consist of brain MRI images depicting brain tumors, Alzheimer's, and multiple sclerosis. These datasets (representing different brain pathologies) are joined to create one cumulative dataset that has been balanced so that all classes comprise approximately the same number of images. This balancing process helps to ensure that the classification accuracy of the hybrid model has not been adversely affected due to a single class being over-represented and thus improving classification accuracy.

The training, validation, and testing subsets of the unified cumulative dataset will be created using either stratified sampling or random sampling methods (to preserve the proportionate representation of the various classes in the unified cumulative dataset), but if there is a known bias in the way that the hybrid model will be trained then stratified sampling should be used. Resized to their respective input image size, the unified cumulative dataset's three subsets will also undergo a standard/process normalization procedure to allow the model to learn using standardized input data.

Moreover, thresholding and morphological operations are employed to create pseudo-mask images for segmentation label approximation that are used to train the segmentation model.

Method: Pseudo masks are created based on thresholding in order to give weak supervision to segmentation. This would reduce the need of manually labelled data and yet enable learning.

U-Net is a model which is trained to learn the mapping between input image and segmentation mask. U-Net encoder-decoder architecture is able to retrieve contextual and spatial information, and this helps to segment brain regions as given in Equation (36).

$$M=g(X) \tag{36}$$

The segmentation mask is used to obtain the region of interest. This process enhances signal-to-noise-ratio because the targeting of relevant brain structures is done only, as defined in Equation (37).

$$XROI = X \odot M \tag{37}$$

The weights of DenseNet201 are pretrained. It is very well connected and can reuse features effectively and has a stronger gradient flow, as represented in Equation (38).

$$F = f_{dense}(X_{ROI}) \tag{38}$$

MRI data are learned with high level disease specific features to be able to fine tune the network thus improving classification

To get feature vectors, the classification head is eliminated. These multidimensional characteristics are rich structural patterns of MRI images, as computed using Equation (39).

$$F_{vec}=\text{Pooling}(F) \tag{39}$$

The features extracted are used to train the LightGBM. The tree growth method of LightGBM is leaf-wise, resulting in LightGBM being more efficient and training in less time without performance loss.

The model minimises categorical cross-entropy loss. This will ensure that the probability is estimated correctly in multi-class classification, as formulated in Equation (40).

$$L = -\sum_{i=1}^{12} y_{i \log(\widehat{y_i})} \tag{40}$$

Output: For each input MRI image, the predicted class label based on the output of the hybrid model is computed. The hybrid model forecasts the likelihood of each class and the class with the highest probability is chosen as the final prediction:

The model can be used to classify multiple kinds of brain diseases into the corresponding category of those diseases, and using segmentation in conjunction with the model provides better classification by ensuring that predictions are limited to clinically significant areas of the brain and thus will result in a higher accuracy for the model's predictions and greater interpretability of the predictions produced by the model.

3.3.7 Grad-CAM(Explainable AI Model)

In order to improve the interpretability of the model and offer visual reasoning to support the predictions of the proposed hybrid deep learning model, Gradient-weighted Class Activation Mapping (Grad-CAM) is employed. In medical image analysis, particularly brain MRI classification, it is not just important to obtain high accuracy of prediction, but also to know the areas of the image that the model used in making the decision. Grad-CAM satisfies this need by rendering visual heatmaps that specify any significant parts that affect the classification result.

Grad-CAM is a gradient-based visualization method that is specifically targeted at convolutional neural networks. In contrast to the conventional interpretability approaches, Grad-CAM uses the internal feature maps of the model and the gradients flowing through them to highlight the spatial parts that are the most important to a specific class prediction. This renders it very appropriate in deep learning models in medical imaging where the localization of abnormalities spatially is critical.

The main benefit of Grad-CAM is that it yields class-discriminative localization, i.e. only the regions that are relevant to predicting a particular class are highlighted using Grad-CAM. This will enable the system to mark areas in a visual form (tumors, lesions or abnormal brain tissues) that affected the prediction. Consequently, Grad-CAM will convert the model into a black-box system into a more understandable and reliable diagnostic system.

Working of Grad-CAM

Grad-CAM works by following a series of steps, which bridge model predictions to spatial features representations:

Input Image and Forward pass: The trained model is fed with a brain MRI image to produce prediction scores on each of the classes. The model uses several convolutional layers with the image, producing feature maps that encode both spatial and semantic data,as defined in Equation (41).

$$Y=f(X) \tag{41}$$

where:

- X= input image

- f = trained model
- y= output prediction scores

Target Layer is selected: Grad-CAM is applied to a particular convolutional layer, usually the final convolutional layer of the network. The reason behind selecting this layer is that it is able to preserve valuable spatial data and at the same time provide high level semantic features that are useful in classification, as expressed in Equation (42).

The target class is selected as:

$$c = \arg max(y) \tag{42}$$

Gradient Computation : The slope of the estimated class score is determined in regard to the feature maps in the chosen layer. These gradients show the relative significance of each feature map in the prediction of the given prediction. When a feature map has a strong effect on the prediction, the values of the gradient will be greater.

Importance Weight Calculation: The gradients are summed up in the spatial dimensions to calculate the importance weights of each feature map. These weights are the contribution of each channel of features in terms of contribution to the final prediction, as formulated in Equation (43).

$$\alpha_k = \frac{1}{H+W} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \tag{43}$$

Combination of Feature Maps with Weights: The importance weights obtained are used to sum up the feature maps to generate a coarse localization map. The areas of the picture that affect the forecast the most are the ones that are marked by this map.

Activation Function : Only the positive contributions are retained with a non-linear activation. This makes sure that only the regions which have a positive impact on the prediction are brought out, and negative effects are pushed aside.

Resizing and Normalization : The resultant activation map is upsampled to the size of the original image and normalized to increase the clarity of the visualization. This enables the heatmap to be properly aligned with the input MRI image as computed using Equation (44).

$$L_{norm} = \frac{L - \min(L)}{\max(L) - \min(L)} \tag{44}$$

Region-Based Masking (Brain Mask Integration): A brain-region mask is used to enhance the interpretation and eliminate the background noise. This step will make sure that the heatmap is confined to the area of interest (brain) and the background areas are suppressed. The brain region is extracted by morphological operations and thresholding techniques.

Generation of Heatmap and Overlay: A color mapping technique is used to transform the normalized activation map into a colored heatmap. This heatmap will then be overlaid on the original MRI picture. The last visualization underlines the most significant areas and does not distort the context of the initial image.

Meaning in the Proposed System: Grad-CAM is used in the proposed hybrid model on the model that performs the best (UNet++ + EfficientNetV2-S + XGBoost pipeline) to provide a visual representation of the classification outcome. Because the system is already doing segmentation to target regions of interest, Grad-CAM also increases interpretability by showing which specific areas (within those segmented) affected the decision.

This two-tier interpretability segmentation-based localization and Grad-CAM-based explanation guarantees that the model does not only achieve an accurate classification, but also offers useful visual information. The regions highlighted may be associated with clinically significant structures like tumor or lesion, which medical professionals can verify the predictions of the model.

Grad-CAM is crucial in converting the proposed system into a framework that is interpretable and clinically sound. It enhances transparency as it visually emphasizes the areas where predictions are made and fosters confidence in the model. This would make the system more applicable in real-world medical scenarios where explainability is critical in decision making and validation.

3.4 Description of Datasets ,Requirements and Tools

The proposed system will make use of several publicly available datasets of brain MRI to create a powerful and general multi-disease classification system. These datasets are meticulously chosen to encompass three big neurological disorders, brain tumors,

Alzheimer disease and multiple sclerosis. The model can be trained on varying imaging patterns, disease stages, and variations in conditions of acquisition, which enhances the applicability of the model to real-world conditions, by using data collected by other sources.

Figure 3.2: Sample MRI Images

The brain tumor dataset is made up of MRI images which have been classified into four classes; Normal, Glioma, Meningioma and Pituitary tumor. These classes are healthy brain states as well as various types of tumors with varying structural features. Tumors of the glial and meningeal type (e.g., gliomas and meningiomas) are healthy brain states. The morphological subtype of glioma tumors is diffuse, heterogeneous in appearance/shape, and involves an infiltrative growth pattern. The morphological subtype of meningioma tumors is defined with well-developed edges, typically superficial to the cerebrum, while pituitary tumors are typically flat (i.e., no protrusion) and located adjacent to the pituitary gland. Variations of the two types of tumor are included within the model for training purposes to allow for recognition of different morphology and pattern of location within the brain, which is important for accurate classification.

The dataset on Alzheimer disease includes MRI images which are classified into four categories, namely, Non-Demented, Very Mild Demented, Mild Demented and Moderate Demented. These groups are progressive levels of the Alzheimer disease; structural changes that occur in the brain become more pronounced as the disease progresses. It is especially difficult to recognize the changes at an early stage because of the slight differences in brain structure (e.g., slight shrinkage or alterations in tissue

density). The dataset is able to learn fine-grained differences, and detect early neurodegeneration because there are several stages involved.

The multiple sclerosis data set contains MRI images that will be classified as Control (healthy) and MS-affected cases, which will be further divided according to imaging views including; axial and sagittal orientation. Multiple sclerosis is also a condition of the brain that is characterized by lesions as a result of damage of the myelin sheath. These lesions differ in size, shape and location hence difficult to detect. Having control and diseased images will aid the model in differentiating normal brain structures and abnormal lesion patterns under varying orientations.

All three datasets are merged to form a single and unified dataset with several classes to enhance the strength of the model. But, as the original datasets might be unevenly distributed in terms of classes, a dataset balancing strategy is used. The classes will be scaled to fit an equal amount of images (1000 images per class in the final set-up). It does so by employing random sampling and duplication, making sure that the model does not get biased towards those classes that have more samples.

Along with balancing, the same preprocessing steps are used in all datasets. These involve scaling of images to a fixed size, normalization of pixel values and standardization of image formats. This preprocessing guarantees consistency of input data that is essential in the consistent training of deep learning models. In addition, augmentation methods are used like rotation, flipping, and shifting to enhance data diversity and better generalization.

The mixed dataset gives a holistic picture of various and diverse brain diseases and the model is able to learn inter-class variance (between diseases) and intra-class variance (within the same disease). This joint dataset incorporation improves the potential of the proposed hybrid model to conduct correct and dependable classification of a broad set of neurological disorders.

On the whole, the utilization of various and balanced datasets is important to enhance the performance, robustness, and the ability to generalize the system. It guarantees that the model is not confined to a particular disease or dataset but can be easily applied to various brain disorders within a single model, which makes it applicable to the real-world medical use.

The execution of the proposed hybrid deep learning system to detect brain diseases involves a systematic integration of the hardware resources, software frameworks, libraries as well as auxiliary tools. As the system consists of several steps including data pre-treatment, data segmentation, feature extraction, classification, and visualization, it is crucial to make sure that the computational environment can efficiently process high-dimensional medical image data and a complex model structure.

Hardware Requirements:

To effectively process big MRI datasets and train deep learning models, the system has to have a high-performance computing environment. The system enabled with a graphics card is strongly suggested since convolutional neural networks and segmentation systems are characterized by a large number of matrix operations and simultaneous calculations. A GPU is utilized in this project, and in this case, it is NVIDIA Tesla P100 or similar that will support the models used in this project, including UNet++, EfficientNetV2-S, and DenseNet201. Use of GPU acceleration can greatly decrease the time spent on training and enable exploration of various models and hyperparameters.

A system with a minimum of 12 -16GB of RAM is necessary to be able to load data, preprocess, and perform batch operations without memory-bottlenecks, in addition to the GPU. The use of high-speed storage (SSD) also can be used to access data faster, particularly with large datasets and intermediate data like augmented images, trained models, and result visualizations. All these hardware components are used to provide a smooth flow of the entire pipeline, which involves training and evaluation.

Software Requirements:

Python is used to develop the system, being popular in machine learning and deep learning because of its simplicity and vast ecosystem. This is implemented in interactive development platforms like Jupyter Notebook and Kaggle Notebook that offer a combined coding platform, visualization, and experimentation platform. These also enable the use of the gpu and make it easier to handle datasets.

Models are developed using deep learning systems like PyTorch, and TensorFlow/Keras. The freedom and flexibility of PyTorch are mostly employed in the

implementation of segmentation models (U-Net and UNet++), as well as hybrid pipelines, based on their dynamic computation graph. In earlier stages, it is built with TensorFlow/Keras to build classification models like EfficientNetB7. These frameworks have intrinsic training, optimization and model examination capabilities, which allow the building of complex architectures efficiently.

Libraries and Tools:

The system uses a number of libraries to support various aspects of the system:

OpenCV (cv2): This is important in image preprocessing which involves resizing, color conversion, thresholding and morphological operations. It is also applied in the creation of pseudo-masks, and image transformations necessary in segmentation.

NumPy: It offers high-performance numerical computing and is popular in processing image arrays and mathematical operations during the pipeline.

Pandas: Pandas are used to create and manipulate structured data, such as file paths, labels and metadata. It eases the process of data management and incorporation among various phases.

matplotlib and seaborn: Visualization of the training performance, such as accuracy plots, loss plots and confusion matrices. Such visualizations can be used to analyze the performance of models and determine the possible improvements.

Scikit-learn: It is applied to split the dataset, evaluation measures, and to produce classification reports. It gives solid measures of evaluation of model performance in terms of accuracy, precision, recall and F1-score.

Segmentation Models PyTorch (SMP): Includes pre-built examples of state-of-the-art segmentation models including U-Net and UNet++, allowing one to easily create segmentation modules.

Torchvision: Loads pre-trained deep learning models (e.g. EfficientNetV2-S and DenseNet201). It also offers image transformation and utilities needed in training the models.

XGBoost: A strong gradient boosting algorithm which is applied in Hybrid Model 1 to classify. It enhances performance through the combination of several decision trees and maximization of prediction accuracy.

LightGBM: A hybrid model 2 uses the LightGBM, a fast and efficient gradient boosting framework. It uses the leaf-wise tree growth approach in order to give improved accuracy and less training time.

TensorFlow ImageDataGenerator: The ImageDataGenerator is utilized in mini projects where the data augmentation process is done through rotation, flipping and scaling among other data transformations.

Visualization and development tools:

Model development and experimentation with the model on cloud-based platforms like Kaggle is used. They are also suitable in large-scale deep learning projects because these platforms offer free access to GPUs, pre-made libraries, and can easily integrate datasets. They also do not require intricate local setup, and can be experimented with more quickly.

Graphviz is a visualization tool that is used to create architectural diagrams of deep learning models. These diagrams are used to comprehend how the data flows in the various layers and are also useful in documentation and presentation. Also, saving and loading trained models tools are applied so as to make them reproducible and deployable in the future.

Workflow Support and System integration.

A combination of these tools allows creating a complete end to end system. The workflow comprises of dataset preparation, preprocessing, segmentation, feature extraction, classification, evaluation and interpretability. Every component is backed by certain tools and libraries so that there is modularity and scalability of the system.

The combination of deep learning systems and machine learning algorithms enables the system to take advantage of both the feature learning and boosting methods. This mixed solution enhances the overall performance and strength. Moreover, the system can work effectively even when large datasets and sophisticated models are involved due to the utilization of a GPU acceleration and libraries optimization.

Overall Significance:

The choice of suitable requirements and tools is critical in the success of the suggested system. Properly configured hardware and software environment will guarantee effective training, correct forecasts and a flawless flow of the pipeline steps. Advanced libraries and frameworks allow implementing state-of-the-art models, and analysis and documentation are provided with the help of visualization and development tools.

On the whole, the selected requirements and tools have a solid base to create a scalable, efficient, and reliable brain disease detection system that will be able to process various datasets and ensure high-performance outcomes in practice.

CHAPTER 4

Results and Observations

4.1 Stepwise description of Results

4.1.1 EfficientNetB7 results

Figure 4.1 shows the graph of accuracy demonstrates the effectiveness of training and validation of the EfficientNetB7 model in 20 epochs. The training accuracy curve (blue) rose steadily to nearly 98 percent in the last epoch starting at approximately 66 percent in the first epoch. The validation (orange) accuracy also rose similarly climaxing at nearly 98 percent after 20 epochs. The difference between training and validation curves is slight which demonstrates the indicators of the model that generalized well on the unseen data with insignificant overfitting. The ultimate assessment yielded a total accuracy of around 98-percent, indicating that the model was efficient in identifying intricate qualities in MRI scanning to differentiate among the eight classes (four stages of Alzheimer and four types of the brain tumors). The close-to-parallel character of the two curves emphasizes the stability of training dynamics, high generalization, and high predictive power of the proposed design.

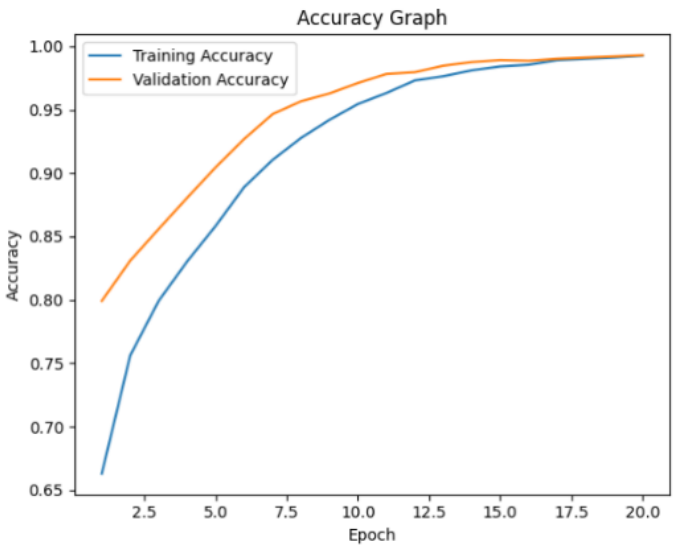

Figure 4.1: Training and Validation Accuracy of EfficientNetB7

Figure 4.2 shows the loss graph shows loss that the values of training and validation loss have reduced in 20 epochs. The training loss (blue curve) shows an initial high at

roughly 1.5 in the first epoch, and its value continuously falls toward zero by around the 20th epoch. Likewise, the validation loss (orange curve) begins at about 0.47 and

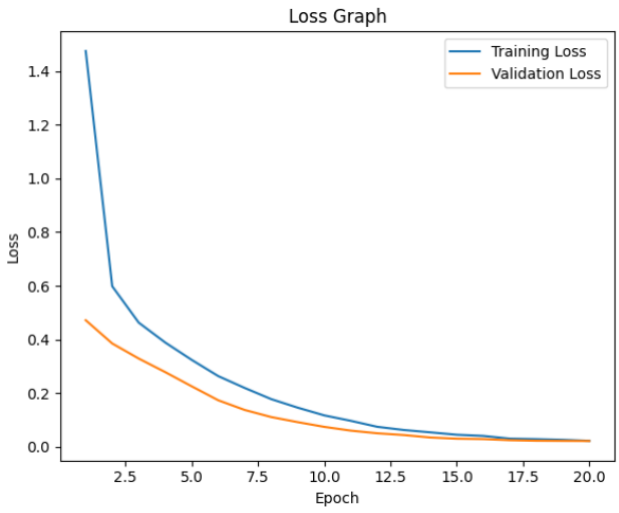

Figure 4.2: Training and validation Loss of EfficientNetB7

exhibits a steady decrease, even approaching zero as it comes to the last epochs. The results clearly help in saying that the model has not overfit as the training and validation loss levels are comparable to each other. The quick reduction of loss in the early epochs proves the effective work of the features learning, and the progress toward the low loss value in later epochs proves the stable training dynamics.

On the whole, the loss graph confirms that the EfficientNetB7 model successfully reduced the number of classification mistakes and showed stability over not only the training but also the validation dataset, which explains high accuracy (up to 98 percent) in the accuracy graph.

The confusion matrix in Figure 4.3 indicates the classification accuracy of the proposed EfficientNetB7 model in eight diagnostic classes: Mild Demented, Moderate Demented, Non-Demented, Very Mild Demented, Glioma Tumor, Meningioma Tumor, Normal Brain and Pituitary Tumor

Categories of Alzheimer's Disease:

The classes Mild Demented (1425/1440), Moderate Demented (1440/1440), and Very Mild Demented (1417/1440) were successfully classified with near-perfect accuracy except occasional misclassification in the adjacent dementia stage.

Non-Demented (1370/1440) performed well too, however there were few cases that were wrongly classified as Mild Demented and Very Mild Demented, which is logical, because of the similarities in visual appearance of the MRI scans.

Categories of brain tumors:

Glioma Tumor (1410/1440) was classified correctly with minimal confusions with Meningioma and Normal Brain.

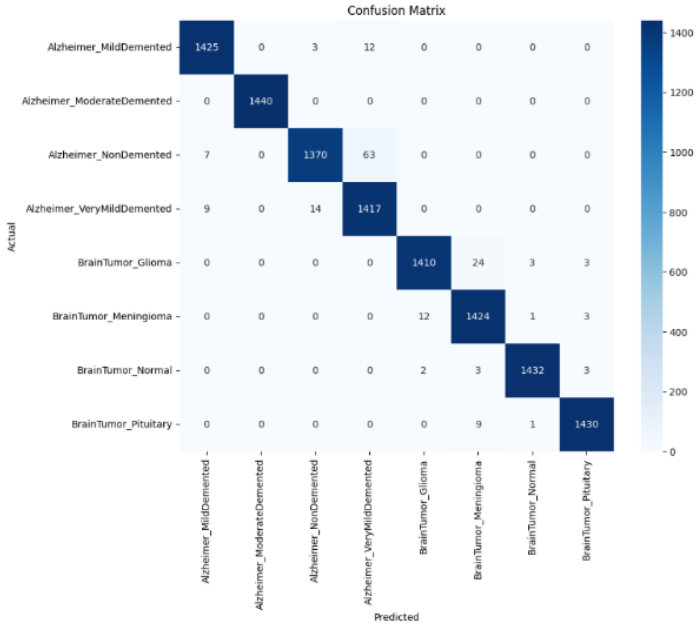

Figure 4.3: Confusion Matrix of EfficientNet B7

Meningioma Tumor (1424/1440) also yielded high recognition, whereas there were only a few misclassifications.

Normal Brain (1432/1440) had a low degree of confusion between tumor classes.

Pituitary Tumor (1430/1440) turned out to be very strong in classification with few misclassification errors

On the whole, the model shows a high discriminative level in all eight classes, and the level of accuracy in most cases (more than 97%). Misclassifications were mostly concentrated between the classes with the similar structural features (e.g., Non-Demented vs. Very Mild Demented, Glioma vs. Meningioma), which is consistent with clinical decision-making issues. This confusion matrix confirms the efficiency of the proposed framework using EfficientNetB7 to deal with a multi-disease, multi-class classification task on MRI scans.

4.1.2.EfficentNet B3 and AlexNet Results

Figure 4.4 represent the graph of accuracy shows how effectively training and validating the EfficientNet-B3 model in 10 epochs. The training accuracy curve goes upwards steadily from the first epoch and reaches about 95% accuracy by the 10th epoch and the effective learning of discriminative MRI features. The validation accuracy curve also shows a similar trend and reaches close to 92% accuracy and also stays aligned closely with the training curve. The small gap between the training and validation curves during the process of training the model imply proper generalization with little over-fitting. The final evaluation is accurate approximately 95% overall thus giving an idea that the model exhibits the capacity to capture intricate patterns in the MRI scan. The nearly parallel nature of the curves indicates the presence of stable training dynamics, impressive generalization to performance, and reliable predictive performance, i.e. across several classes of brain diseases.

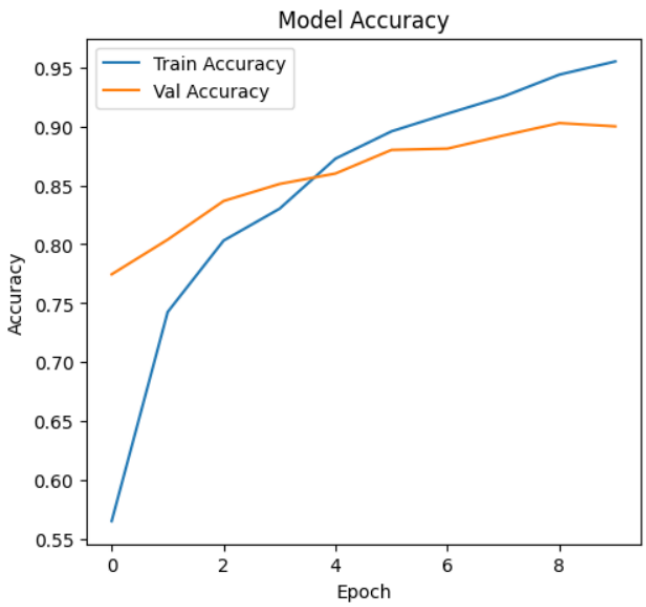

Figure 4.4: Training and validation accuracy of EfficientNetB3

Figure 4.5 The loss curve is a plot of the training and validation performance for the EfficientNet-B3 model for 10 epochs. The training loss drops drastically at the beginning stages of epochs and continues to decline steadily and shows the effective optimization and learning of MRIs features. The validation loss also shows a gradual decrease and closely tracks the training loss curve indicating stable learning behavior. The fact that training and validation loss have a small gap indicates there was little

overfitting in the training process.

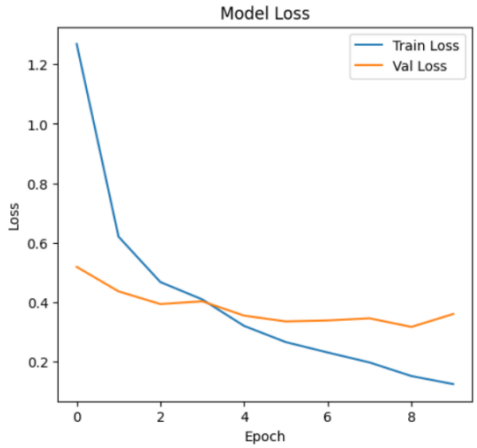

Figure 4.5: Training and validation Loss of EfficientNetB3

Minor changes in validation loss over subsequent epochs will signal a regular adjustments in the model and not an instability. Overall, the loss curves demonstrate smooth convergence behavior, stable training process and robust generalization of the proposed model on unseen MRI data.

Figure 4.6 shows the confusion matrix of the EfficientNet-B3 model showing its performance on all 12 disease classes, Brain Tumor types, stages of Alzheimer, and Multiple Sclerosis types. High values of most classes in the diagonal cells means that most of the input MRI images are classified correctly by the model.

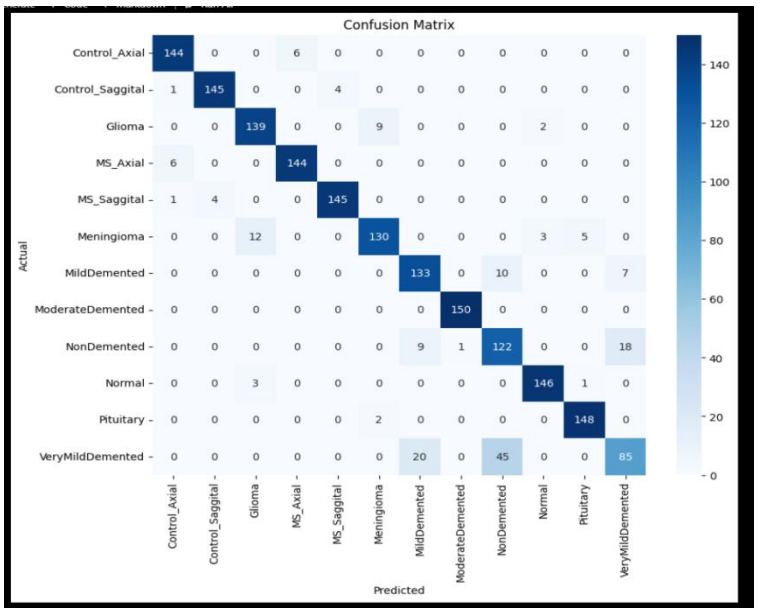

Figure 4.6: Confusion Matrix of EfficientNet B3

The model exhibits great performance in classifying MRI orientations and disease types Control_Axial, Control_Saggital, Glioma, MS_Axial, MS_Saggital, Meningioma, Normal and Pituitary with counts of over 130 correct predictions. These powerful diagonal arrangements are an indication of EfficientNet-B3 inability to capture fine-grained structural and texture-based features in MRI scans.

In the case of Alzheimer, the model is able to perform well in the Mild Demented, Moderate Demented, and Very Mild Demented, however a small number of misclassifications are found among the adjacent mental stages. This is anticipated as there are some minor variations in the structural brain alterations throughout the early development of Alzheimer. As an example, VeryMildDemented is confused with NonDemented and MildDemented because of the similarity in the clinical pattern of early stages. The confusion matrix also indicates that misclassifications between entirely unrelated disease categories are minimal which evidences the high discriminative capability. There are very few brain tumor images that are being reclassified as Alzheimer or MS and vice versa implying that the model is very robust in multi-disease classification. All in all, the EfficientNet-B3 model is characterized by a high degree of accuracy and reliability, with high predictions on all classes with limited confusion in structurally similar or clinically overlapping classes. This attests to the ability of the model to be a useful diagnostic instrument to multi-disease brain MRI analysis.

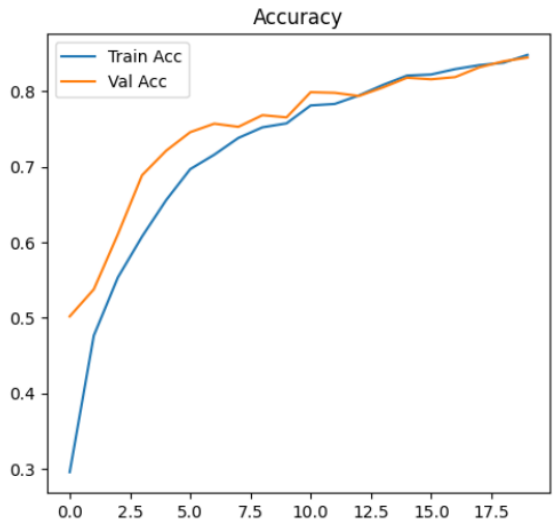

Figure 4.7: Training and validation accuracy of AlexNet

Figure 4.7 represents the plot of the accuracy curve gives the training and validation

curve of the AlexNet model for the training epochs. Training accuracy incrementally from the first pixels, and achieves high accuracy ~ 83% towards the last epoch, indicating fruitful feature learning from MRI image. The validation accuracy shows a similar trend and is close to the training curve which reflects a reasonable generalization with limited overfitting. Minor deviations between the curves are noticed for intermediate epochs, which are normal for CNN training. The accuracy attained at the end, which is about 83%, shows that AlexNet works well enough as a baseline model. However, the relatively lower accuracy comes as a highlight of its drawback when compared with the more advanced EfficientNet-B3

Figure 4.8 represents the loss curve shows the training and validation performance of AlexNet model in the training epochs. The training loss sharp decreases during the first several epochs and keeps decreasing slowly, which points to the good optimization and learning of MRI features

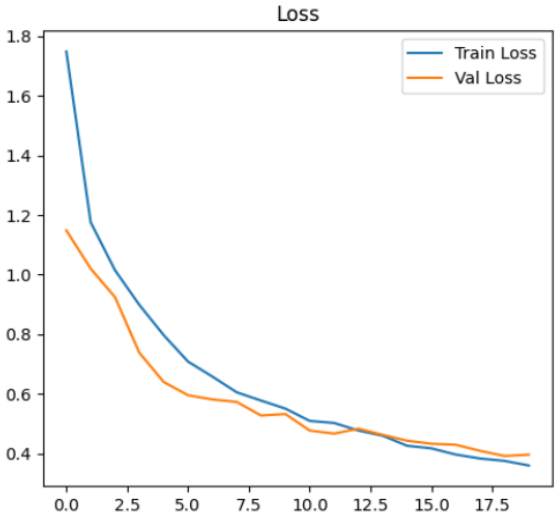

Figure 4.8: Training and validation Loss of AlexNet

The validation loss also follows a similar decreasing trend and stays near the training loss during the training process. The slight difference between the two curves is revealing limited overfitting and constant learning behavior. Minor fluctuation in validation loss in later epochs is seen, which is typical, during the training of deep learning models. Overall, the loss curves prove smooth convergence and reliable performance of the AlexNet model, though it falls short in its last loss comparing it to EfficientNet-B3.

The confusion matrix in Figure 4.9 illustrates that AlexNet has a reasonable accuracy in a few classes and specifically: Control_Axial, Control_Saggital, MS_Axial,

MS_Saggital, Meningioma, ModerateDemented, Normal and Pituitary classes where the majority of the predictions lie along the diagonal. It means that the model is able to accurately detect categories which have greater structural variations in MRI scans.

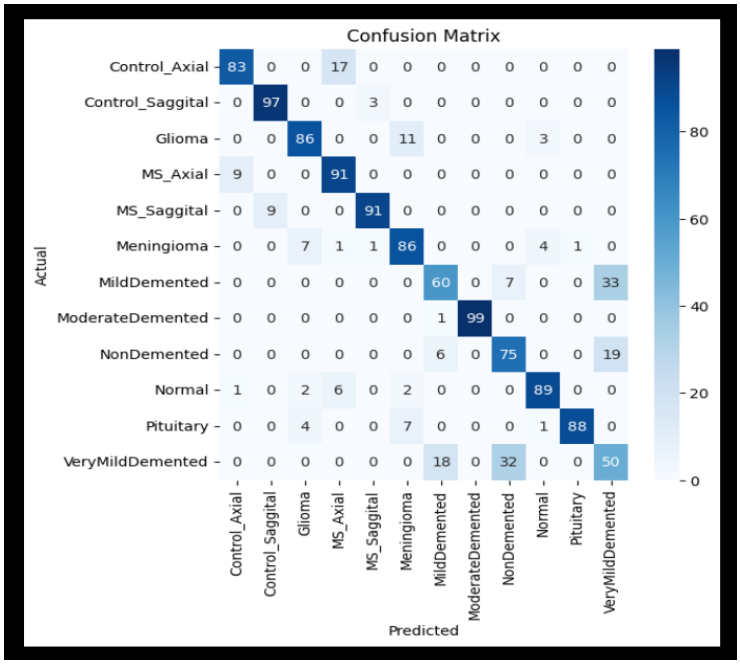

Figure 4.9: Confusion Matrix of AlexNet

However, the model has difficulties with classes of Alzheimer-related like NonDemented, VeryMildDemented and MildDemented, in which slight structural differences cause apparent misclassifications. The shallow architecture of AlexNet is less able to capture the early-stage features of Alzheimer and this causes overlaps between the adjacent cognitive stages.

All in all, the matrix demonstrates that AlexNet is effective when it comes to the distinctly defined MRI classes but fails to differentiate the subtle ones, which once again proves that more sophisticated architectures such as EfficientNet-B3 can be used to achieve more reliable classifications of multiple diseases.

The Table 4.1 shows the comparison of the class-wise performance between AlexNet and EfficientNet-B3 using the precision, recall, and F1-score metrics. EfficientNet-B3 also performs consistent better on higher scores in most brain tumor, Alzheimer's disease and multiple sclerosis classes. On the whole, the result from the above

comparison shows the superiority of EfficientNet-B3 over AlexNet for multi-brain disease classification.

Table 4.1: Class-wise comparison of AlexNet and EfficientNet-B3 performance

Class	AlexNet precision	AlexNet Recall	AlexNet F1 score	Efficient Net-B3 precision	Efficient Net-B3 Recall	Efficient Net-B3 F1 score
Control _Axial	0.99	0.83	0.86	0.95	0.96	0.95
Control _Saggtial	0.92	0.97	0.94	0.97	0.97	0.97
MS_Axial	0.79	0.91	0.85	0.96	0.96	0.96
MS_ Saggtial	0.96	0.91	0.93	0.97	0.97	0.97
Glioma	0.87	0.86	0.86	0.90	0.93	0.91
Pitutary	0.99	0.88	0.93	0.96	0.99	0.97
Meningioma	0.81	0.86	0.83	0.97	0.87	0.89
Normal	0.92	0.89	0.90	0.97	0.97	0.97
Moderate Demented	1.0	0.99	0.99	0.99	1.0	1.0
Non Demented	0.66	0.75	0.70	0.69	0.81	0.75
Very Mild Demented	0.49	0.50	0.50	0.77	0.57	0.65
Mild Demented	0.71	0.60	0.65	0.82	0.89	0.89

Figure 4.10 depicts the accuracy curve for the proposed EfficientNet V2-S, where changes in training and validation accuracies with respect to the number of epochs are illustrated. As it can be noticed, training accuracy increases continuously, which shows that the model learns discriminatory information successfully from the MRI image dataset. Likewise, the validation accuracy curve is increasing as well, and both curves are very close to each other.

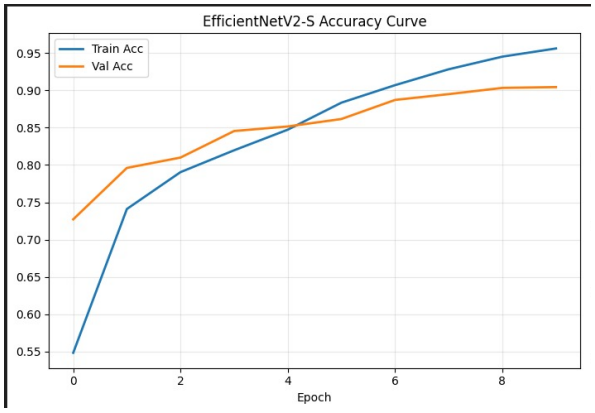

Figure 4.10: Training and validation accuracy of Hybrid Model-1

This reveals that there is little overfitting in the proposed hybrid model. Consequently, this means that the model performs quite well in terms of training and generalizing its learning capacity.

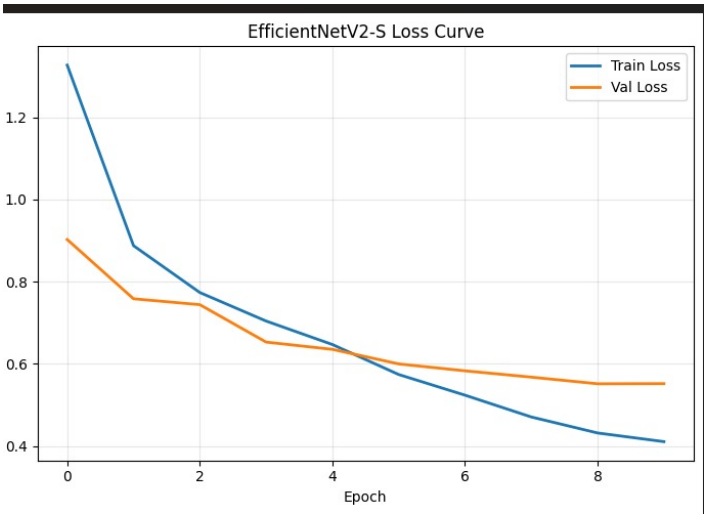

Figure 4.11: Training and validation Loss of Hybrid Model-1

Figure 4.11 above is an illustration of the loss curve for the proposed EfficientNet V2-S, which shows how the training and validation losses change during the process of epochs. As can be seen from the graph, the training loss curve is consistently falling, suggesting that the proposed model successfully optimizes prediction errors. Validation loss decreases gradually too; it is clear from Figure 4.12 that the trends are quite similar, showing that the model performs well and generalizes well on unseen data. In conclusion, it is important to note that the model is not prone to overfitting since training and validation losses follow each other quite closely.

The confusion matrix demonstrates how the hybrid model can be used to predict 12 classes, with the values in the diagonal corresponding to correct predictions, and the off-diagonal values being the cases of misclassifications. The high accuracy of the model is supported by the high diagonal pattern that confirms the ability of the model to differentiate the various diseases of the brain.

Moderate Demented, Control (Axial and Sagittal), MS (Axial and Sagittal), and Pituitary have very high correct predictions, which suggests a robust feature learning.

Other classes of tumors such as Glioma and Meningioma are also able to work with few errors.

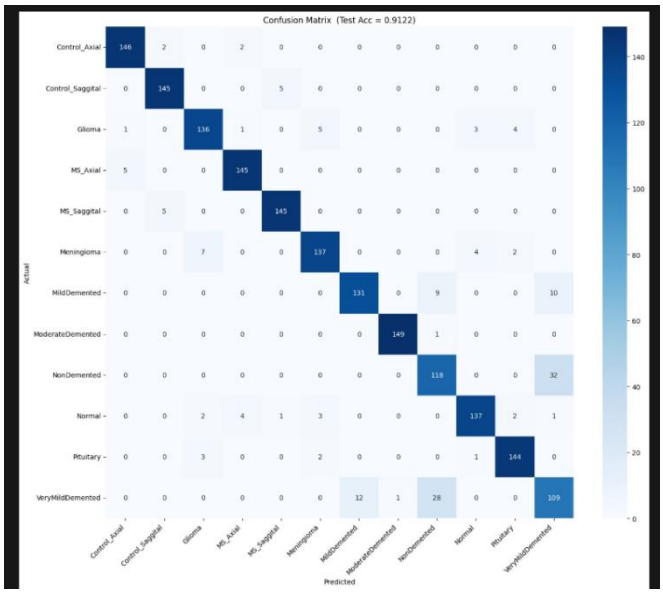

Figure 4.12: Confusion Matrix of Hybrid Model 1

Other misclassifications are within similar classes, particularly between Non-Demented and Very Mild Demented, as there are minor variations in MRI patterns. All in all, the model has high reliability, and the error tends to be in visually similar categories

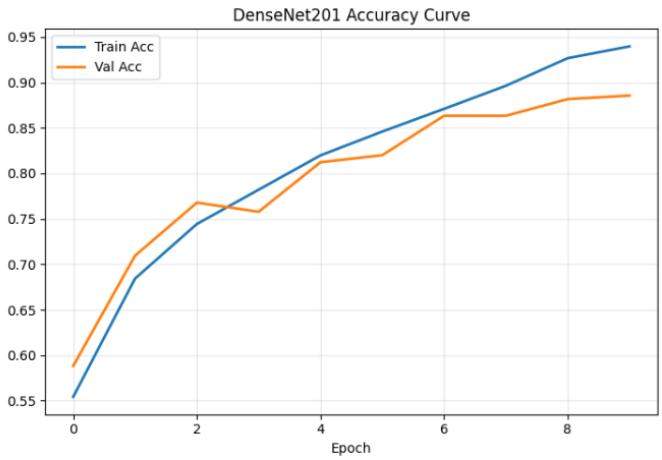

Figure 4.13: Training and validation accuracy of Hybrid Model-2

The accuracy curve for the second hybrid model (Figure 4.13) shows how the accuracy of both the training and validation data continue to increase as the number of epochs increases, which indicates that the model is able to learn the characteristics of the MRI images effectively. While the training and validation accuracies of this model increase

at a similar rate, the gap, or difference, between these two accuracies is greater than the gap between these two accuracies with the first hybrid model, indicating that this model is exhibiting relatively poor generalization performance. Nevertheless, there is also a clear upward trend in the validation accuracy, which suggests that the model is learning in a stable manner. Overall, the accuracy curve of the second hybrid model demonstrates strong performance overall, although not quite as much as the accuracy curve of the first hybrid model exhibits.

The loss curves of the DenseNet 201, both training and validation losses, across epochs are represented in Figure 4.14. The data in this figure indicates that the training loss is continuously decreasing; therefore it can be concluded that the hybrid model is appropriately learning to minimize prediction error. The validation loss is showing a decreasing trend, although it remains slightly greater than the training loss across the course of the training process, indicating that there exists a slight difference between the performance between training and validation, and that the ability of this hybrid model to generalize is comparatively poorer than the first hybrid model. In summary, both training and validation losses exhibited a gradual decrease over time demonstrating that there was stable convergence; however, this hybrid model is slightly less consistent when compared to the original hybrid model

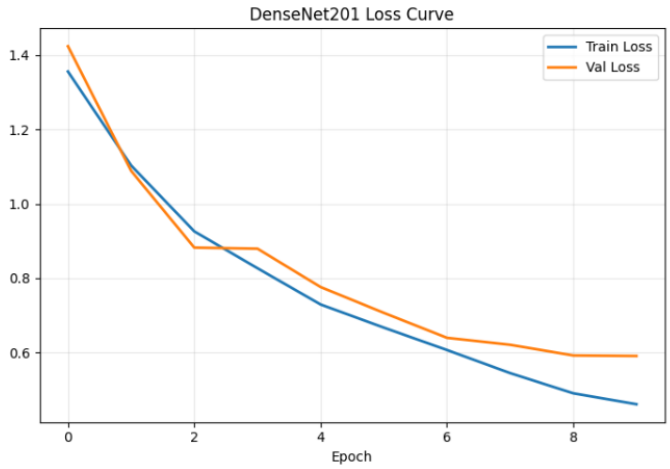

Figure 4.14: Training and validation Loss of Hybrid Model-2

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The below confusion matrix is an indication of the second hybrid model performance, which has a test accuracy of approximately 88.56. The correct prediction indicated by the diagonal values is high and its high occurrence demonstrates that the model is good

in most of the 12 classes.

There are a number of classes that are extremely accurate: Moderate Demented (150 correct predictions), Control Axial (145) and Control Sagittal (145) suggesting that the model learns very clear and distinguishable features in these classes. As well, the performance of classes such as Normal (140) and MS Axial (142) are also strong.

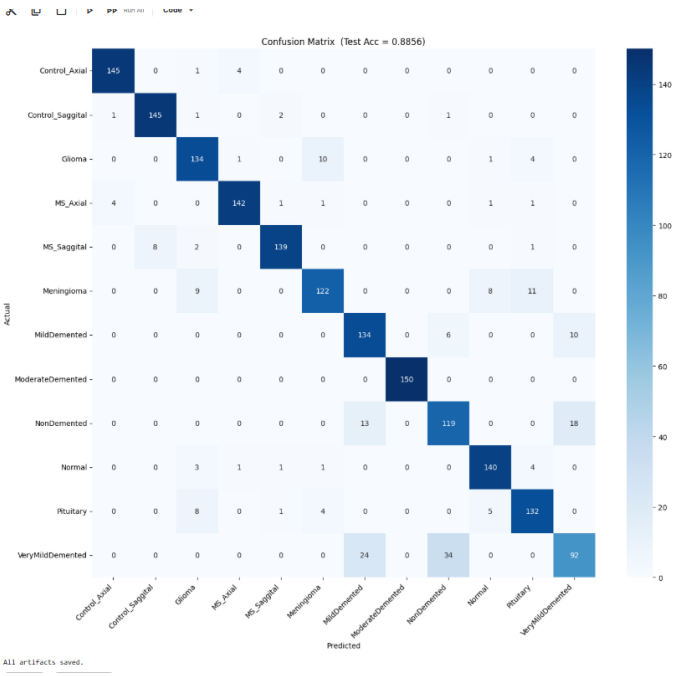

Figure 4.15: Confusion Matrix of Hybrid Model-2

But this model, compared to the earlier confusion matrix indicates a few more misclassifications. It is worth noting that there is a lack of clarity between Mild Demented, Non-Demented and Very Mild Demented, with such overlap in the initial stages of Alzheimer, making it more difficult to classify. Also, a few mistakes are observed between Glioma and Meningioma, Pituitary and other classes of tumors, because of the resemblances in the structure of the tumors.

On the whole, despite the good performance of the model and the stable classification, the fact that the misclassification increases in comparison to the first hybrid model shows that the generalization ability is slightly worse. The analysis below justifies this conclusion: Hybrid Model 1 can be used to solve the complex multi-class brain disease

classification tasks better in comparison to Hybrid Model 2.

Table 4.2: Class-wise comparison of hybrid models performance

Class	Hybrid Model-1 precision	Hybrid Model-1 Recall	Hybrid Model-1 F1 score	Hybrid Model-2 Recall	Hybrid Model-2 precision	Hybrid Model-2 F1 score
Control _Axial	0.96	0.97	0.96	0.96	0.96	0.96
Control _Saggtial	0.95	0.96	0.96	0.94	0.94	0.95
MS_Axial	0.95	0.96	0.96	0.94	0.95	0.95
MS_ Saggtial	0.96	0.96	0.96	0.92	0.96	0.94
Glioma	0.91	0.90	0.91	0.89	0.84	0.87
Pitutary	0.94	0.96	0.95	0.88	0.99	0.87
Meningioma	0.93	0.91	0.92	0.81	0.88	0.84
Normal	0.94	0.91	0.92	0.93	0.91	0.91
Moderate Demented	0.99	0.99	0.99	1.0	1.0	1.0
Non Demented	0.75	0.78	0.77	0.79	0.74	0.76
Very Mild Demented	0.71	0.72	0.72	0.61	0.76	0.61
Mild Demented	0.91	0.87	0.89	0.89	0.78	0.83

The table 4.2 shows a comparison of two hybrid models across multiple classes of brain disease, as evaluated through precision, recall and F1-score. Hybrid Model-1 achieves uniform and strong performance within classes, scoring near perfect for both Control and many other categories including MS as well as Moderate Demented. Similarly, Hybrid Model-2 achieves state-of-the-art performances in some classes (Moderate Demented) but shows a substantial drop for Meningioma, Glioma and Very Mild Demented categories. In general, Hybrid Model-1 seems to have higher stability and generalization across multiple classes due to it having learned features that perform well across difficult or imbalanced categories. Both models also tend to perform poorer in Non-Demented and Very Mild Demented classes, indicating that these are the more difficult categories to classify. In total, compared to Hybrid Model-2 captures data for

consistency and reliability from the sense of all dataset metrics outdone by the Hybrid Model-1.

4.2Result Analysis and Test Case Results

The proposed system was evaluated experimentally based on numerous deep learning and hybrid models in order to examine their effectiveness on the multi-class classification of brain diseases. The findings make it very clear that model architecture, the ability to extract features, and the integration of segmentation is a great contributor of the overall performance.

The Best model with the highest performance in Mini Project 1 was the EfficientNetB7 with an approximation of 98-99 accuracy. The training and validation accuracy curves exhibit a single smoothly increasing trend and the loss curves exhibit a single smoothly decreasing trend to minimal values. High generalization and minimal overfitting can be demonstrated by the near parallelness between the training and validation curves. This performance is also confirmed by the classification report, with the majority of classes having a precision, recall and F1-scores of nearly 1.0. This means that EfficientNetB7 can easily bring out finer spatial details of MRI images, thus very useful in the ability to differentiate between stages of Alzheimer and the types of brain tumors.

In the Mini Project 2, the overall accuracy of EfficientNetB3 was approximately 95%, with a good performance and a consistent convergence. The accuracy curve, loss curve show that the model is learning well and that there is a small difference between training and validation, indicating good generalization. Conversely, AlexNet was only able to attain an accuracy of around 83%. Despite the stable training behavior of AlexNet, its comparatively shallow architecture restricts its capability to detect sophisticated hierarchical characteristics of MRI images. This is especially in difficult classes like Mild Demented and Very Mild Demented where nuances need more detailed representation of features. The relative performance of the modern architectures as well as the traditional CNNs is evidently revealed in the comparative results where modern architectures such as EfficientNet outperform AlexNet.

The Hybrid Model 1 (UNet++ + EfficientNetV2-S + XGBoost) has a test accuracy of about 91.22 and good and balanced performance on all 12 classes. The confusion matrix shows a preponderance of a diagonal pattern which means that most of the predictions

are accurate. Control Axial, Control Sagittal, MS Axial, MS Sagittal, and Moderate Demented classes demonstrate almost perfect classification, which shows the usefulness of the model in acquiring specific features. UNet++ integration is important as it isolates the region of interest and eliminates the irrelevant background information, thereby enhancing features. The EfficientNetV2-S also provides a more effective extraction of features and classifies the image by learning more complicated decision boundaries, unlike XGBoost, which focuses on the low-level features. The minor misclassifications have been noted to be mainly between Non-Demented and Very Mild Demented, and between Glioma and Meningioma, as they appear visually similar in MRI scans.

Hybrid Model 2 (U-Net + DenseNet201 + LightGBM) also obtained a test accuracy of about 88.56; this is a little lower than Hybrid Model 1. Even though this model demonstrates good performance with well-distinct classes like Moderate Demented, Control classes and Normal brain images, it displays greater confusion in more complicated classes. The confusion matrix indicates that there are significant misclassifications between Mild Demented, Non-Demented and Very Mild Demented, and tumor classes such as Glioma and Meningioma. Although DenseNet201 is an efficient reuse of features due to dense connections and LightGBM is an efficient classifier, the overall discrimination of features is a bit lower than EfficientNetV2-S and XGBoost. Also, the difference between training and validation curves is a little higher which implies a relatively worse generalization.

A comparison between the two hybrid models, in the form of classes, further indicates that Hybrid Model 1 is superior. It yields greater accuracy, recall, and F1-scores in most of the classes, particularly in hard classes that include early-stage Alzheimer disease and tumor differentiation. Hybrid Model 2, in spite of its effectiveness, is less effective in such demanding situations because it has overlapping features.

All in all, the analysis of results proves that the deep-lying architectures and hybrid frames have a considerable enhancement of the classification performance. EfficientNetB7 is the most accurate of all the single models, which affirms its capability to extract features well. Hybrid Model 1 is the most successful hybrid method in terms of accuracy, robustness, and generalization because it combines effectively segmentation, advanced deep learning, and boosting methods. Hybrid Model 2 also

does a good job, but it is not as effective in dealing with more complicated and visually similar classes. These results prove the effectiveness of the combination of segmentation and potent feature extractors and machine learning classifiers as a dependable method of detecting brain diseases with high accuracy and scalability when MRI images are used.

4.2.1TestCase Results of EfficientNetB7

The Figure 4.16 shows the prediction result interface of the designed MRI disease detection system. After uploading an MRI image, it is processed by the model and the predicted disease type is provided together with a confidence score. The scan uploaded has been classified as Alzheimer s Moderate Demented with 99 percent confidence. Prediction result interface of the proposed MRI disease detection system, The system also shows the probability distribution of classes of the fixed eight categories of diagnostics (Mild Demented, Moderate Demented, Non-Demented, Very Mild Demented, Glioma Tumor, Meningioma Tumor, Pituitary Tumor, and Normal Brain).

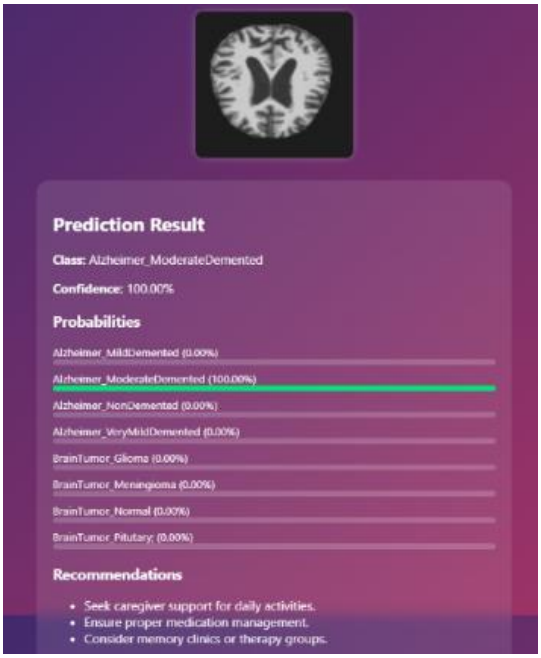

Figure 4.16: Prediction Result of EfficientNetB7 as Alzheimer

This offers clarity in prediction where the user can see how sure the model is in the differentiations of the possible outcomes. Besides, the interface offers clinical suggestions on the identified disorder, including caregiver support, medication, and therapy choices. This combination of diagnostic prediction with materially helpful recommendations makes the tool more practically useful in the use by a healthcare

professional and caregivers alike as a decision-support tool.

The Figure 4.17 shows the results of the developed system prediction brain tumor classification. MRI scan uploaded by the user has been recognized as a case of Brain Tumor-Meningioma with confidence of 100 %. The interface gives the probability distribution of all diagnostic categories where it is shown that the highest probability percentage of 99 is assigned to the meningioma category and the other probability distribution percentages are zero. Besides classification, the system provides recommendations that are applicable to the diagnosis. In meningioma cases, these present options are to observe the tumor, assess the possibility of surgical or radiations modalities in the event of a need and follow-up with routine MRI scans to track activity. The integration of disease classification with practical advice increases the utility of the system as a decision-modeling tool in the clinical setting

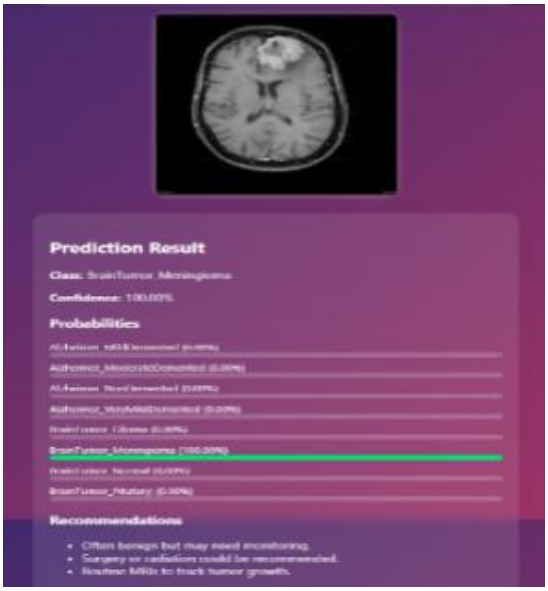

Figure 4.17: Prediction Result of EfficientNetB7 as Tumor

4.2.1TestCase Results of EfficientNetB3 with LIME

Figure 4.18 represents the LIME's visual explanation points to the specific areas of the MRI image that influenced the model to make its prediction of a glioma brain tumor the most. Highlighted in yellow are the areas of abnormal tissue within the brain, in other words, the model is trying to target clinically important tumor regions, and not background structures. This reaffirms that the EfficientNet-B3 model learns relevant spatial features that are related to glioma features.

The close match of the highlighted areas and visible tumor mass leads to improved trust in the decision-making process of the model. LIME thus makes models transparent by giving additional intuitive visual justifications for predictions. Overall, this explanation is conducive to the reliability interpretability of the proposed system to brain tumor.

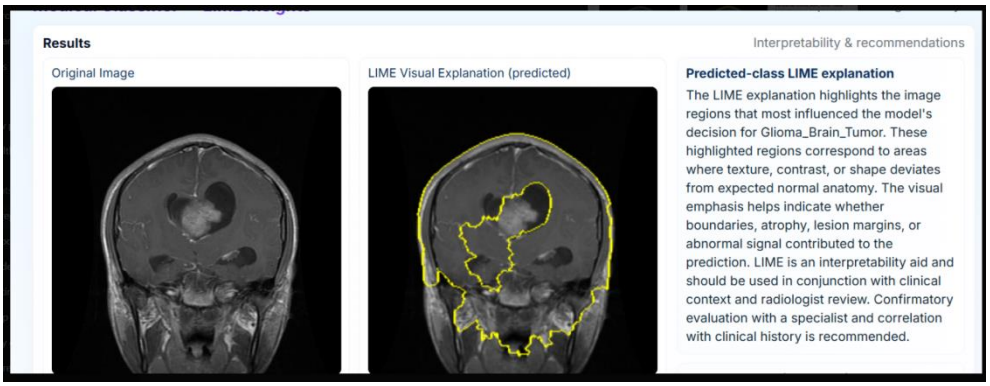

Figure 4.18: Prediction Result of EfficientNet B3 along with LIME Visualisation

The Figure 4.19 presents the calculated risk score, indicating the model's confidence and estimated severity for the predicted condition. It also displays the top five predicted classes, helping users understand alternative possibilities in the diagnosis. Additionally, the interface provides nearby hospital links in Vijayawada, enabling quick access to medical consultation based on the prediction

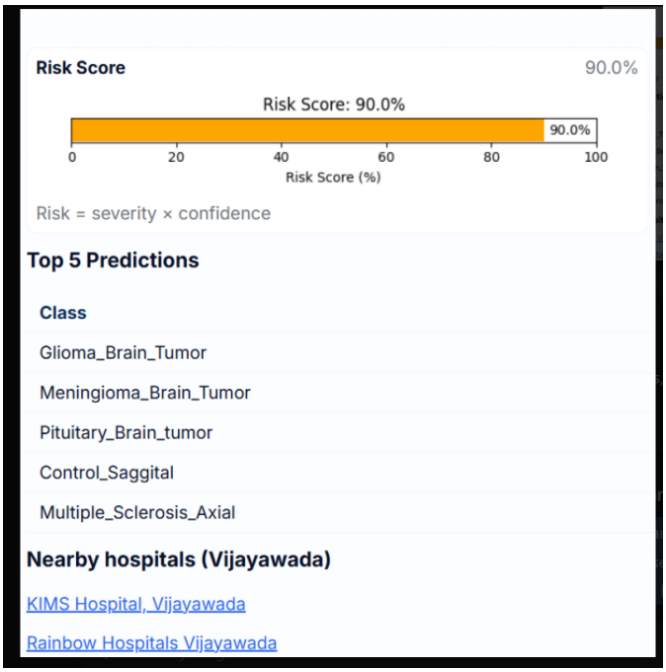

Figure 4.19: Risk Score and Top Predictions of EfficientNet B3

4.2.3 Test Case Results of Hybrid Model-1 with Grad-CAM

Figure 4.20: Result of Hybrid Model -1 along with Grad-CAM as Tumor

The Grad-CAM visualization illustrated in Figure 4.20 depicts the regions of the MRI image that most influence the proposed hybrid model's prediction. The activation map uses a colour intensity scale to highlight the most important areas, where warmer colours such as red and yellow indicate a greater level of influence than cooler colours, which represent areas that have a smaller degree of importance.One can see that this model has identified important brain regions that are associated with the presence of abnormality, indicating that the model has the capability of identifying disease-specific patterns correctly. The regions of interest that are highlighted in the Grad-CAM visualization correspond to critical structures represented on the MRI scan, confirming the model is predicting from meaningful, interpretable features as opposed to irrelevant background information. The use of the Grad-CAM 64isualization therefore enhances the transparency of the proposed hybrid model by providing an understanding of how it has made its decisions, thus improving trust and reliability in the use of the model in actual medical diagnostic settings. The model detected or predicted the image as the glioma tumor with 98 percent confidence.

The Figure 4.21 shows the prediction and interpretability outcome of the proposed hybrid model on a sample MRI image which has a high confidence of 99.86 and is classified as Moderate Demented

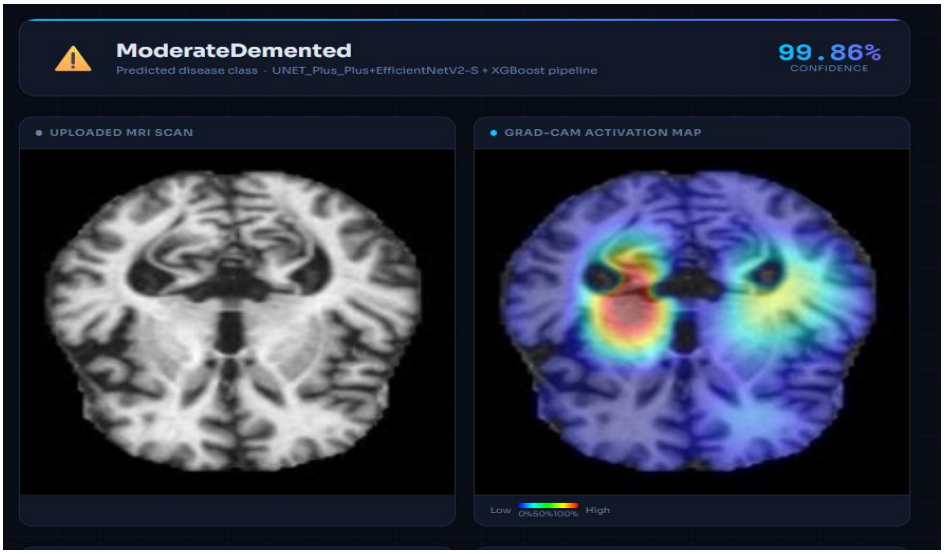

Figure 4.21: Result of Hybrid Model -1 along with Grad-CAM

The image on the left demonstrates the original MRI scan and the one on the right demonstrates the respective Grad-CAM activation map that points out the regions that played the greatest role in the prediction made by the model.

In the visualization, it is clear that the emphasized areas (in red and yellow) are clustered in certain parts of the brain, meaning that the model is drawing attention to the meaningful parts of the brain, and not the irrelevant background areas. This proves that the model is not just making random predictions but it is learning clinically relevant features related to the progression of Alzheimer disease.

Grad-CAM map offers a valuable degree of transparency through which it displays the manner in which the model makes its decision. The strength of the heatmap shows the significance of various regions, with the warmer colors representing a greater contribution to the classification. Thanks to the apparent localization of activated regions, the model can be supposed to capture the patterns of diseases like the alteration of brain tissues.

On the whole, this outcome indicates a high predictive accuracy and the interpretability of the proposed system. The good classification results and visualization explain why the model is more reliable and can be applied to medical practice since clinicians will have an easier time interpreting and trusting the automated predictions.

4.3Observations from Work

The proposed system was tested in an experiment using many deep learning and hybrid models to test their performance regarding the multi-classification of brain diseases. The results illustrate clearly that model architecture, the feature extraction capability and the combination of segmentation is a significant contributor of the total performance.

The EfficientNetB7 with an approximation of 98-99 accuracy was the best model that performed the best in Mini Project 1. The training and validation accuracy curve have a smooth increasing trend and the loss curves have the smooth decreasing trend to minimum values. The near-parallels between the training curve and the validation curve can attest to high levels of generalization and low levels of overfitting. The classification report also confirms this performance, most of the classes having a precision, recall and F1-score of almost 1.0. This implies that EfficientNetB7 would readily elicit finer spatial points of MRI images, hence highly valuable in the possibility of distinguishing between stages of Alzheimer and the kind of brain tumors.

In the Mini Project 2, the general accuracy of EfficientNetB3 was about 95, and a good performance and a stable convergence was observed. Accuracy curve, loss curve reveal that the model is learning properly and that there is not too much difference between training and validation which is an indication of good generalization. Conversely, AlexNet was only able to attain an accuracy of around 83%. Although AlexNet is stable in its training behavior, its rather shallow architecture limits its ability to identify deep hierarchical features of MRI images. This is particularly in challenging classes such as Mild Demented and Very Mild Demented where features require more finer details of depiction. The relative performance of the modern architectures as well as the traditional CNNs is evidently revealed in the comparative results where modern architectures such as EfficientNet outperform AlexNet.

The Hybrid Model 1 (UNet++ + EfficientNetV2-S + XGBoost) with its test accuracy of approximately 91.22 has good and balanced performance on all 12 classes. The confusion matrix indicates that there is a dominance of a diagonal shape indicating that most of the predictions are correct. Control Axial, Control Sagittal, MS Axial, MS Sagittal and Moderate Demented classes are almost perfectly classified which

demonstrates the utility of the model in obtaining specific features. UNet++ integration is important as it isolates the region of interest and eliminates the irrelevant background information, thereby enhancing features. The EfficientNetV2-S also offers a better extraction of features, and classifies the image by learning more complex decision boundaries than XGBoost that concentrates on the low-level features. The slight misclassifications have been identified as largely between Non-Demented and Very Mild Demented and between Glioma and Meningioma since they are similar in appearance in MRI scans.

Hybrid Model 2 (U-Net + DenseNet201 + LightGBM) got a test accuracy of approximately 88.56; slightly less than Hybrid Model 1. Although this model has proven to work well with classes that are clearly differentiated such as the Moderate Demented, Control classes and Normal brain images, it shows more of confusion in the more complex classes. The confusion matrix shows that, the case of Mild Demented, Non-Demented and Very Mild Demented is greatly misclassified with the tumor Glioma and Meningioma. Although DenseNet201 is an efficient reuse of features due to dense connections and LightGBM is an efficient classifier, the overall discrimination of features is a bit lower than EfficientNetV2-S and XGBoost. Also, the difference between training and validation curves is a little higher which implies a relatively worse generalization.

An analysis between the two hybrid models, in the form of classes, further reveals that Hybrid Model 1 is better. It is more accurate, recalls and F1-scores on most of the classes, especially the hard classes which involve early-stage Alzheimer disease and tumor differentiation. Hybrid Model 2, despite its effectiveness, is not as effective in such challenging situations since it shares features.

All in all, the analysis of results proves that the deep-lying architectures and hybrid frames have a considerable enhancement of the classification performance. EfficientNetB7 is the most accurate of all the single models, which affirms its capability to extract features well. Hybrid Model 1 is the most effective hybrid approach in terms of accuracy, robustness and generalization since it unites well segmentation, sophisticated deep learning, and boosting technique. Hybrid Model 2 also is not as effective with dealing with more complicated and visually similar classes, but it does a good job. These findings confirm the efficiency of the combination of segmentation

and powerful feature extractors and machine learning classifiers as a reliable way of identifying brain diseases with high accuracy and scalability in case MRI images are employed.

CHAPTER-5

Conclusion and Future work

5.1 Conclusion

The current project introduces a smart and detailed system of brain disease diagnosis which covers such brain disorders as brain tumors, Alzheimer disease, and multiple sclerosis through the employment of MRI images. The experiment manages to combine both deep learning and hybrid machine learning methods to deal with the problem of multi-class medical image classification.

The experimental findings prove that state-of-the-art deep learning architectures are very helpful at identifying sophisticated features on MRI scans. The best performing of the individual models was EfficientNetB7, which had a high accuracy and good generalization. A comparison between EfficientNetB3 and AlexNet further proves that contemporary architectures perform much better than traditional convolutional networks when it comes to processing complex medical data.

The hybrid models proposed complement the system by integrating segmentation, feature extraction and classification methods. U-Net and U-Net++, which are used to extract the region-of-interest, enhance the quality of input features, since it eliminates background noise. The combination of UNet++ + EfficientNetV2-S + XGBoost showed the best results in terms of accuracy and generalization, being better than the second hybrid model. This proves the point that by incorporating the algorithms of boosting with the deep learning characteristics, the classification results could be greatly enhanced.

The findings also demonstrate the relevance of preprocessing methods like balancing, normalization, and augmentation of datasets in the attainment of stable training and minimization of overfitting. Even with the high overall performance, misclassifications in visually similar classes were found, especially at earlier stages of the Alzheimer disease as well as with some types of tumors, which indicates the complexity inherent in the medical image analysis.

Moreover, interpretability methods like Grad-CAM can also be added to the system to increase its transparency by showing predictions of the model in a visual format. This is of great essence especially in medical applications where it is important to know the rationale behind the predictions to develop trust.

To sum up, the suggested system shows that hybrid deep learning models can offer a powerful, precise, and scalable approach to multi-disease brain MRI classification. The combination of segmentation, sophisticated feature extractor, and machine learning classifiers would greatly enhance performance and render the system acceptable in real-life clinical decision support scenarios.

5.2Future Study

To enhance the generalization and robustness of the proposed work, larger and more diverse datasets with multiple sources can be used. This would be even better with the inclusion of 3D MRI data rather than 2D images which will give volumetric data of the brain structures.

The segmentation phase can be improved upon in place of pseudo-masks, fully annotated datasets are used to extract the region-of-interest more precisely. Future research can also consider the more advanced hybrid and ensemble methods to enhance the performance of classification.

Moreover, accuracy and interpretability can be further improved by incorporating transformer-based models and other explainable AI methods. To make the system more practical and useful in clinical practice, it is possible to develop a real-time application and integrate MRI data with clinical information.

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Appendix

The appendix provides a technical description of the key stages involved in the proposed hybrid deep learning pipeline for multi-class brain disease detection. Each stage plays a crucial role in improving accuracy, robustness, and interpretability of the system for medical diagnosis using MRI images.

Stage 1: Dataset Integration and Balancing

Primary Logic: data_preprocessing.py

The system begins by combining multiple datasets related to brain tumor, Alzheimer's disease, and multiple sclerosis into a unified dataset. A balancing strategy is applied to ensure that each class contains an equal number of samples (1000 images per class), thereby reducing class imbalance and improving model generalization.

```
IMG_EXTS      = (".jpg", ".jpeg", ".png", ".bmp", ".tif", ".tiff")
combined_root = "/kaggle/working/balanced_dataset"
os.makedirs(combined_root, exist_ok=True)

def balance_and_copy(src_dict):
    for label, src_path in src_dict.items():
        dst_path = os.path.join(combined_root, label)
        os.makedirs(dst_path, exist_ok=True)
        files = [f for f in os.listdir(src_path) if f.lower().endswith(IMG_EXTS)]
        if not files:
            print(f"[WARNING] No images in {label} at {src_path}")
            continue
        random.shuffle(files)
        if len(files) >= TARGET_COUNT:
            selected = files[:TARGET_COUNT]
        else:
            extra    = np.random.choice(files, size=TARGET_COUNT - len(files), replace=True)
            selected = list(files) + list(extra)
        for i, f in enumerate(selected):
            shutil.copy(os.path.join(src_path, f),
                        os.path.join(dst_path, f"{i}_{f}"))
        print(f"  {label}: {len(selected)} images")

print("Balancing datasets...")
balance_and_copy(brain_tumor_paths)
balance_and_copy(alz_paths)
balance_and_copy(ms_paths)
print("Dataset balanced.\n")
```

Figure A.1 : Dataset Balancing

Stage 2: Pseudo Mask Generation for Segmentation

Primary Logic: preprocessing.py

Since manual annotations are unavailable, pseudo masks are generated using image processing techniques such as Otsu thresholding and morphological operations. These masks act as weak supervision signals for training the segmentation model.

```
def create_pseudo_mask(rgb_img):
    """
    Otsu-threshold pseudo-mask — used ONLY as supervision signal
    while training the UNet++.  Not used during classification.
    """
    gray = cv2.GaussianBlur(cv2.cvtColor(rgb_img, cv2.COLOR_RGB2GRAY), (5, 5), 0)
    _, mask = cv2.threshold(gray, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU)
    if np.mean(mask > 0) > 0.6:
        mask = 255 - mask
    kernel = np.ones((5, 5), np.uint8)
    mask   = cv2.morphologyEx(mask, cv2.MORPH_OPEN,  kernel)
    mask   = cv2.morphologyEx(mask, cv2.MORPH_CLOSE, kernel)
    n_lbl, lbl, stats, _ = cv2.connectedComponentsWithStats(mask, connectivity=8)
    if n_lbl > 1:
        largest = 1 + np.argmax(stats[1:, cv2.CC_STAT_AREA])
        mask = np.where(lbl == largest, 255, 0).astype(np.uint8)
    return (mask > 0).astype(np.float32)
```

Figure A.2 : Mask generation

Stage 3: Segmentation using U-Net++

Primary Logic: segmentation.py

The U-Net++ architecture is used to segment regions of interest (ROI) from MRI images. Dense skip connections improve feature propagation and boundary detection.

```
seg_model = smp.UnetPlusPlus(
    encoder_name="efficientnet-b2",
    encoder_weights="imagenet",
    in_channels=3,
    classes=1,
    decoder_attention_type="scse",
).to(device)
```

Figure A.3 : Segmentation Model

```python
def preprocess_single_image(image_path, seg_model_cpu):
    img_u8 = read_rgb_image(image_path)
    mask   = unetpp_predict_mask(seg_model_cpu, img_u8)
    img_f  = img_u8.astype(np.float32) / 255.0 * np.expand_dims(mask, -1)
    return _normalize(torch.tensor(img_f).permute(2, 0, 1)).unsqueeze(0)
```

Figure A.4 : ROI Extraction

Stage 4: Deep Feature Extraction using EfficientNetV2-S

Primary Logic: feature_extraction.py

EfficientNetV2-S is used as a feature extractor to learn high-level representations from segmented MRI images. Pretrained weights from ImageNet improve learning efficiency.

```python
in_features = effv2_model.classifier[1].in_features
effv2_model.classifier = nn.Sequential(
    nn.Dropout(p=0.2, inplace=True),
    nn.Linear(in_features, num_classes),
)
effv2_model = effv2_model.to(device)

cls_loss_fn = nn.CrossEntropyLoss(label_smoothing=0.05)
```

Figure A.5 : Model Definition

```python
feature_extractor = copy.deepcopy(effv2_model)
feature_extractor.classifier = nn.Identity()   # expose 1280-d pool output
feature_extractor = feature_extractor.to(device)
feature_extractor.eval()
for p in feature_extractor.parameters():
    p.requires_grad_(False)
torch.save(feature_extractor.state_dict(), "/kaggle/working/efficientnetv2s_feature_extractor.pth")
print("Feature extractor ready (1280-d embeddings).\n")
```

Figure A.6 : Feature Extraction

This stage converts images into meaningful feature vectors capturing complex patterns.

Stage 5: Classification using XGBoost

Primary Logic: classifier.py

The extracted feature vectors are passed to the XGBoost classifier, which uses gradient boosting to improve classification accuracy and reduce overfitting.

```
print("Training XGBoost...")
xgb_model = xgb.train(
    params=xgb_params,
    dtrain=dtrain,
    num_boost_round=350,
    evals=[(dtrain, "train"), (dval, "val")],
    evals_result=evals_result,
    early_stopping_rounds=30,
    verbose_eval=50,
)

xgb_model.save_model("/kaggle/working/xgboost_brain_multiclass.json")
with open("/kaggle/working/xgboost_brain_multiclass.pkl", "wb") as f:
    pickle.dump(xgb_model, f)

print("XGBoost model saved.\n")
```

Figure A.7 : XGBoost Training

XGBoost enhances performance by combining multiple weak learners into a strong predictive model.

Stage 6: Model Evaluation and Performance Analysis

Primary Logic: evaluation.py

The performance of the hybrid model is evaluated using standard metrics such as accuracy, classification report, and confusion matrix.

```
y_prob   = xgb_model.predict(dtest)
y_pred   = np.argmax(y_prob, axis=1)
test_acc = accuracy_score(y_test, y_pred)

print(f"Test Accuracy: {test_acc:.4f}\n")
```

Figure A.8 : Evaluation

This stage validates the effectiveness of the model across multiple disease classes.

Stage7: Grad-CAM Heatmap Generation

Primary Logic: gradcam.py

Grad-CAM is used to generate visual explanations by highlighting important regions in MRI images that influence the model's prediction. It utilizes gradients and feature maps from the target convolutional layer to produce a heatmap.

```
model = model.to(device)
model.eval()

gradcam = GradCAM(model, target_layer)
cam = gradcam(input_tensor, class_index=class_index)
gradcam.remove_hooks()
h, w = original_image.shape[:2]
cam_resized = cv2.resize(cam, (w, h), interpolation=cv2.INTER_CUBIC)
cam_resized = np.clip(cam_resized, 0, 1)
brain_mask = compute_brain_mask(original_image)  # float32, [0..1], shape (H, W)
cam_masked = cam_resized * brain_mask             # suppress background
if cam_masked.max() > 1e-8:
    cam_masked = cam_masked / cam_masked.max()
heatmap_uint8 = np.uint8(255 * cam_masked)
heatmap_colored = cv2.applyColorMap(heatmap_uint8, cv2.COLORMAP_JET)  # BGR
original_bgr = cv2.cvtColor(original_image, cv2.COLOR_RGB2BGR)
mask_3ch = np.stack([brain_mask] * 3, axis=-1)   # (H, W, 3)
blended_brain = cv2.addWeighted(original_bgr, 0.55, heatmap_colored, 0.45, 0)
overlay = (blended_brain * mask_3ch + original_bgr * (1.0 - mask_3ch)).astype(np.uint8)
```

Figure A.9 : CAM generation

This step produces a class activation map that emphasizes disease-relevant regions such as tumors, lesions, or abnormal brain tissues.

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